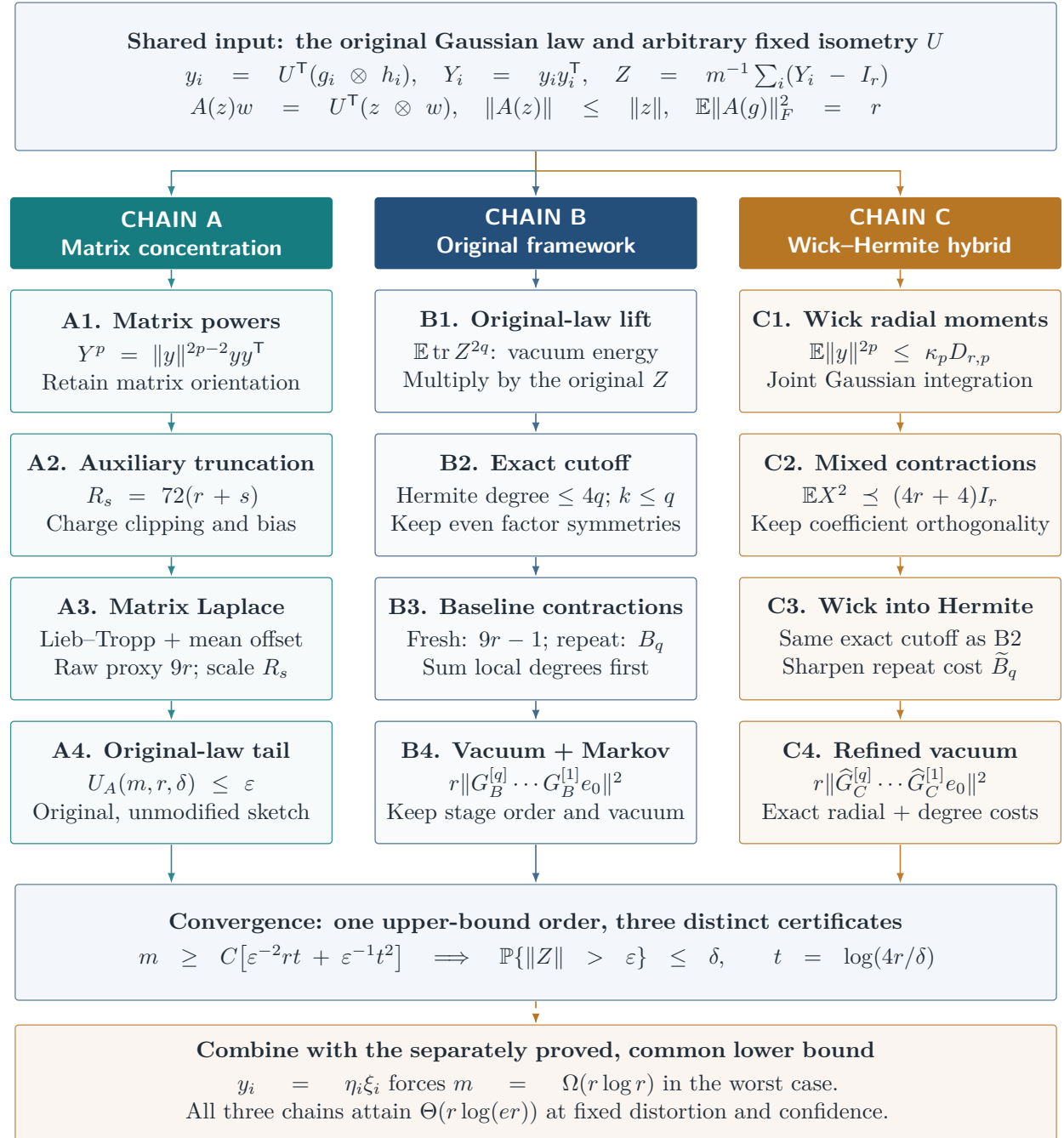


Sharp Gaussian Khatri–Rao embeddings

Three proof chains: where they split and meet

Read downward. A and B retain the original concentration and framework proofs. C is the separate Wick–Hermite hybrid. The model and final sample-size order are shared; the intermediate estimates and constants are not.



Here $\kappa_p = (2p-1)!!$ and $D_{r,p} = \prod_{\ell=0}^{p-1} (r+2\ell)$. Page 2 fixes the A/B/C comparison. The Wick-assisted concentration variant is documented separately, not substituted into A or labelled C. All application-specific proofs follow in this document.

Three-chain comparison

Same Khatri–Rao problem; three visibly separate proof routes.

The main labels are fixed throughout: **A = original matrix concentration**, **B = original framework**, and **C = combined Wick–Hermite framework**. C has its own local estimates, proof section, stage matrices, and numerical certificate.

Comparison	Chain A (§3)	Chain B (§4)	Chain C (§5)
Retained object	One-row matrix powers $\mathbb{E}Y^p$.	Original-law vacuum walks, sample supports, and local degrees.	Exact Wick radial/mixed moments inserted into the same Hermite walks.
Domain	An auxiliary rare-outcome truncation; restore the original law.	Exact cutoff: total degree $\leq 4q$, active count $\leq q$, even symmetries.	The same exact cutoff. No outcome truncation and no independent replacement.
Gaussian input	Collective Lipschitz row estimate and directional fourth moments.	The same baseline row estimate; Hölder and hypercontractivity at local degree d .	Sharp radial product $\kappa_p D_{r,p}$ and orthogonal mixed-fourth contractions.
Variance used	Raw second-moment proxy $9r$.	Centered activation proxy $9r - 1$.	Centered activation proxy $4r + 4$.
Assembly	Lieb–Tropp matrix Laplace with a deterministic mean offset.	Positive sector majorant and ordered vacuum propagation, then Markov.	The same propagation, with smaller costs and refined active-degree budgets.
Explicit output	Tail allowance U_A in (3.12).	Moment and tail certificates in (4.24).	Refined moment certificate (5.27).
What is not claimed	An optimal concentration constant.	Necessity of the framework for this problem.	A new sketch, a new sample-size order, or a pure Wick-only proof.

Explicit sufficient row counts: A versus B versus C

$\varepsilon = 1/2$, $\delta = 0.01$; identical model and subspace scope. B uses its original stage-dependent certificate; C uses the radial/degree refinement.

r	A: concentration	B: framework	C: Wick–Hermite	$q_B = q_C$
10	91 000	60 000	12 800	3
100	580 000	472 000	111 000	5
1 000	6 200 000	5 150 000	1 250 000	7
10 000	72 000 000	61 000 000	15 000 000	8

Table 1: The three main certificates. These are selected sufficient theorem bounds, not empirical row counts or global minima. Parameters and interval checks appear in Section 7 and Appendix C.

What this comparison means

All three prove the same rate. The change from B to C measures the Wick–Hermite refinement while keeping the final moment order fixed. It does not establish universal superiority over concentration: the separate Wick-assisted variant A^W also improves A, and is compared with C in Appendix D. It is not a fourth main chain.

1 Model, theorem, and comparison at a glance

1.1 The unchanged sketch and its quantifiers

Let $n_1, n_2, r, m \geq 1$ be integers with $r \leq n_1 n_2$, and fix a deterministic isometry

$$U : \mathbb{R}^r \longrightarrow \mathbb{R}^{n_1} \otimes \mathbb{R}^{n_2}, \quad U^\top U = I_r.$$

For $1 \leq i \leq m$, let $g_i \sim N(0, I_{n_1})$ and $h_i \sim N(0, I_{n_2})$; all samples and both factors are mutually independent. Define

$$S = \frac{1}{\sqrt{m}} \begin{pmatrix} (g_1 \otimes h_1)^\top \\ \vdots \\ (g_m \otimes h_m)^\top \end{pmatrix}, \quad y_i = U^\top (g_i \otimes h_i), \quad (1.1)$$

$$Y_i = y_i y_i^\top, \quad X_i = Y_i - I_r, \quad Z = U^\top S^\top S U - I_r = \frac{1}{m} \sum_{i=1}^m X_i.$$

Independence gives $\mathbb{E}(g_i \otimes h_i)(g_i \otimes h_i)^\top = I_{n_1 n_2}$, so $\mathbb{E}Y_i = I_r$ and $\mathbb{E}X_i = 0$. The event $\|Z\| \leq \varepsilon$ is exactly the squared-norm embedding guarantee

$$(1 - \varepsilon)\|x\|^2 \leq \|S U x\|^2 \leq (1 + \varepsilon)\|x\|^2 \quad \text{for every } x \in \mathbb{R}^r.$$

Every probability statement below is for *each fixed* U , with constants uniform in U, n_1, n_2 . It is not a simultaneous statement for all subspaces after S is drawn. All logarithms are natural; unmarked matrix norms are operator norms.

Theorem 1.1 (Common endpoint). *There is a universal $C > 0$ such that, for $0 < \varepsilon, \delta < 1$ and $t = \log(4r/\delta)$,*

$$m \geq C \left(\frac{rt}{\varepsilon^2} + \frac{t^2}{\varepsilon} \right) \implies \mathbb{P}\{\|Z\| > \varepsilon\} \leq \delta. \quad (1.2)$$

Each of Chains A, B, and C below establishes this upper bound. A and B use the baseline local estimates; C replaces them by its Wick contractions and then uses the common original-law Hermite interface. Conversely, for $U e_a = e_a \otimes e_1$ with $n_1 = r, n_2 = 2$, there is $c > 0$ such that uniformly over integers $1 \leq m \leq cr \log r$,

$$\mathbb{P}\{\|Z\| > 1/2\} \longrightarrow 1 \quad (r \longrightarrow \infty).$$

Therefore the worst-case sample order at distortion 1/2 and success probability 0.99 is $\Theta(r \log(er))$.

The statement does not assert optimal joint dependence on ε and δ . The upper bound is proved for arbitrary subspace coefficients; only the lower-bound example uses a common factor.

1.2 Reading guide and chain conventions

Section 2 fixes the original local estimates for A and B. Sections 3, 4, and 5 are the three proof chains. C reuses B's exact multiplication/cutoff identities, but supplies its own Wick contractions, repeated-visit budgets, and refined propagation theorem. It is not a renaming of B. The lower bound is shared in Section 6.

Section 7 compares fixed A/B/C certificates. The sharper concentration variant A^W is separated in Appendix D. Classical Gaussian inputs are derived in Appendix A; Lieb's trace concavity is stated and attributed in Section 3.3. No earlier project note is needed for an application-specific proof step.

1.3 Related work and attribution

The sketch is not a new construction. Tensor random projections were studied by Sun, Guo, Tropp, and Udell [2], and fixed-vector polynomial-kernel sketching bounds appear in Ahle et al. [3, Appendix A.2, Theorem 42]. In particular, the order of the $r = 1$ sample bound is already present in that fixed-vector literature; the target here is uniformity over all directions of an arbitrary fixed r -dimensional subspace. Bujanović and Kressner [4] study norm and trace estimation with random rank-one vectors. Bujanović, Grubišić, Kressner, and Lam [5] prove an earlier OSE theorem for this two-factor Gaussian model.

Beretta–Musco [8, Theorem 1.3] give $O(\varepsilon^{-2}r \log^3(r/(\varepsilon\delta)))$ rows in the Gaussian order-two specialization. Their theorem covers more distributions and tensor orders. The comparison here is restricted to that specialization, with an absolute rescaling of distortion when passing between norm and squared-norm conventions. Neither the present constants nor the numerical table are advertised as universal improvements over their broader result.

A version-sensitive predecessor. Saibaba, Verma, and Ballard’s first version [6, Theorem 11] contains a near-linear, polylogarithmic OSE claim and an accompanying proof attempt. The second version [7] omits that assertion. Beretta–Musco’s acknowledgments [8] report that a proof error prompted its removal and credit the earlier strategy. We retain this historical credit but do not use the removed assertion as a theorem. The paper as a whole is not described as withdrawn.

Methodological antecedents. Chain A follows the established truncation-and-concentration architecture for Khatri–Rao sketches; compare [8, Section 2.3]. Its specific calculations retain one-row matrix moments and the distinction between the truncation height and the exponential-moment scale. The matrix Laplace mechanism is the Lieb–Tropp method, not a new concentration theorem [15, 16]. The order of the shared projected-row estimate also follows from the Hilbert-space specialization of Adamczak–Łatała–Meller [13, Corollary 12]; Section 2 supplies an explicit normalization and a short direct proof. Coefficient-sensitive matrix-chaos estimates and Khatri–Rao applications also precede this note in Bandeira, Lucca, Nizić–Nikolac, and van Handel [20]. Their full-matrix norm estimates are not identified with the centered, arbitrary-fixed-subspace covariance theorem proved here.

Chain B is organized using the supplied framework [1]. Its multiplication representation, Hermite decomposition, and Jacobi moment interpretation have classical antecedents [17, 18, 19]. The application-specific content to assess is the sequence of local estimates, the resulting sample bound, and its explicit vacuum certificates, rather than ownership of these representation principles. Section 8.1 separates the sources and roles. No publication-priority claim is made.

The hybrid is separated from both baselines. Chain C combines the classical Wick–Isserlis pairing rule with the Hermite and energy interface of B [14, 21, 1]. The radialization mechanism also appears in the supplied attributed rMPS comparison [22]; that manuscript is cited for internal provenance, not as a theorem about the present Khatri–Rao model. The application-specific refinements were recorded in the supplied Wick supplement [23] and are proved here in Section 5. The rMPS comparison is used as a layout reference only: its reconstruction model, inverse-Gram expansions, and numerical tables are not transferred to the present theorem.

2 Shared baseline coefficient estimates

The same subspace geometry feeds all three chains. This section fixes the pre-Wick estimates used by A and B; C supplies sharper local inputs separately. Keep the map $A(z)$ collected before taking norms: its operator-norm Lipschitz constant is 1, not $\sqrt{r}$. Then keep the rank-one orientation when a matrix-valued estimate is needed.

2.1 Gaussian moment control with explicit normalization

Write γ_N for standard Gaussian measure on $\mathbb{R}^N$ and $\text{Ent}(F) = \mathbb{E}[F \log F] - (\mathbb{E}F) \log \mathbb{E}F$ for $F \geq 0$. The Gaussian logarithmic Sobolev inequality is

$$\text{Ent}_{\gamma_N}(f^2) \leq 2\mathbb{E}_{\gamma_N} \|\nabla f\|^2. \quad (2.1)$$

For a Hilbert-valued homogeneous Hermite polynomial F_d of total degree d , Gaussian hypercontractivity gives

$$\|F_d\|_{L_u} \leq (u-1)^{d/2} \|F_d\|_{L_2}, \quad u \geq 2. \quad (2.2)$$

The normalization and short semigroup proofs appear in Appendix A; these are classical Nelson–Gross inequalities [9, 10, 11]. Differentiating moments under a logarithmic Sobolev inequality is also a standard route to moment estimates; see Aida–Stroock [12]. The following lemma fixes the Gaussian normalization used throughout.

Lemma 2.1 (Lipschitz moments). *For a nonnegative L -Lipschitz function f on a finite-dimensional standard Gaussian space and any real $p \geq 2$,*

$$\|f\|_{L_p}^2 \leq \|f\|_{L_2}^2 + (p-2)L^2. \quad (2.3)$$

Proof. Assume first that f is smooth, bounded, and bounded below by a positive constant, and set $M(p) = \mathbb{E}f^p$.

1. Differentiate the running moment in its exponent.

Since $\frac{d}{dp} \mathbb{E}f^p = \mathbb{E}f^p \log f$, a direct computation gives

$$\frac{d}{dp} M(p)^{2/p} = \frac{2}{p^2} M(p)^{2/p-1} \text{Ent}(f^p).$$

2. Bound the entropy by the logarithmic Sobolev inequality.

Apply (2.1) to $f^{p/2}$ and use $\|\nabla f\| \leq L$:

$$\text{Ent}(f^p) \leq \frac{p^2 L^2}{2} \mathbb{E}f^{p-2}.$$

Hölder’s inequality gives $\mathbb{E}f^{p-2} \leq M(p)^{(p-2)/p}$, so the two previous displays combine into

$$\frac{d}{dp} M(p)^{2/p} \leq L^2 M(p)^{2/p-1} M(p)^{1-2/p} = L^2.$$

3. Integrate from 2 to p .

This yields $M(p)^{2/p} \leq M(2) + (p-2)L^2$, which is (2.3). Smoothing, a positive regularization, and truncation of the scalar test function justify the same bound for Lipschitz f by passage to the limit. $\square$

2.2 The collective conditional Gaussian map

Reshape the columns of U as matrices $B_1, \dots, B_r \in \mathbb{R}^{n_1 \times n_2}$. Then

$$\mathrm{tr}(B_a^\top B_b) = \delta_{ab}, \quad y_a = g^\top B_a h. \quad (2.4)$$

For $z \in \mathbb{R}^{n_1}$ define $A(z) : \mathbb{R}^{n_2} \rightarrow \mathbb{R}^r$ by

$$A(z)w = U^\top(z \otimes w), \quad y = A(g)h. \quad (2.5)$$

Because $U^\top$ is a contraction,

$$\|A(z)\| \leq \|z\|, \quad \mathbb{E}\|A(g)\|_F^2 = r, \quad \|A(z)\|_F^2 \leq r\|z\|^2. \quad (2.6)$$

The second assertion follows by summing $\|B_a\|_F^2 = 1$; the third follows from $\sum_a \|B_a^\top z\|^2 \leq \sum_a \|B_a\|_F^2 \|z\|^2$. Linearity of A makes $z \mapsto \|A(z)\|$ 1-Lipschitz and $z \mapsto \|A(z)\|_F$ $\sqrt{r}$ -Lipschitz.

Lemma 2.2 (Projected-row moments). *For every real $p \geq 2$,*

$$\boxed{\| \|y\| \|_{L_p}^2 \leq r(2p-3) + (p-2)^2.} \quad (2.7)$$

In particular $\| \|y\| \|_{L_p} \leq \sqrt{2rp} + p$.

Proof. 1. Bound the moments of the two Lipschitz norms of $A(g)$.

By (2.6), $z \mapsto \|A(z)\|_F$ is $\sqrt{r}$ -Lipschitz with squared L_2 norm r , and $z \mapsto \|A(z)\|$ is 1-Lipschitz with squared L_2 norm at most r . Lemma 2.1 gives

$$\| \|A(g)\|_F \|_{L_p}^2 \leq r(p-1), \quad \| \|A(g)\| \|_{L_p}^2 \leq r + p - 2.$$

2. Condition on g and integrate over h .

For fixed g , the function $h \mapsto \|A(g)h\|$ has Lipschitz constant $\|A(g)\|$ and squared L_2 norm $\|A(g)\|_F^2$. Lemma 2.1, applied in the variable h alone, gives

$$(\mathbb{E}_h \|A(g)h\|^p)^{2/p} \leq \|A(g)\|_F^2 + (p-2)\|A(g)\|^2.$$

3. Combine the two factor bounds.

Take $L_{p/2}$ norms in g on both sides. Minkowski's inequality in $L_{p/2}$ separates the two summands on the right, and Step 1 bounds each of them:

$$\| \|y\| \|_{L_p}^2 \leq r(p-1) + (p-2)(r + p - 2) = r(2p-3) + (p-2)^2.$$

All coefficients remain collected. No common Schmidt basis and no independence of the coordinates of y is assumed. $\square$

Relation to Banach-space Gaussian chaos bounds. The order $\| \|y\| \|_{L_p} \lesssim \sqrt{rp} + p$ is already a consequence of Adamczak–Latała–Meller [13, Corollary 12], with their function-space exponent set to 2. To check the specialization, set $c_{ij} = U^\top(e_i \otimes e_j)$ and encode the bilinear form using the symmetric coefficient array with cross blocks $c_{ij}/2$ and zero diagonal blocks, acting on the concatenated Gaussian (g, h) . The four coefficient quantities in that corollary are bounded, in their displayed order, by $\sqrt{r/2}$, $\sqrt{r}/2$, $1/\sqrt{2}$, and $1/2$. This gives the stated order. Lemma 2.2 supplies the explicit bound (2.7), not a claim that this moment order is new. The cited corollary is an antecedent for the shared local estimate, not by itself the centered sample-covariance theorem.

2.3 Directional fourth moments and matrix variance

The normal fourth-moment calculation below is a special case of Isserlis’s product-moment formula [14], written out to fix the constant 9. For the related rank-one probing model, see also [4].

Lemma 2.3 (Fourth moments). *For every $x \in \mathbb{R}^r$,*

$$\mathbb{E}\langle x, y \rangle^4 \leq 9\|x\|^4. \quad (2.8)$$

Consequently, with $Y = yy^\top$ and $X = Y - I_r$,

$$\mathbb{E}Y^2 \preceq 9rI_r, \quad \mathbb{E}X^2 \preceq v_r I_r, \quad v_r := 9r - 1. \quad (2.9)$$

Proof. 1. The directional fourth moment.

For unit x , put $B_x = \sum_a x_a B_a$. Equation (2.4) gives $\|B_x\|_F = 1$. Conditional on g , $g^\top B_x h$ is Gaussian with variance $g^\top Q g$, where $Q = B_x B_x^\top \succeq 0$ and $\text{tr } Q = 1$. Diagonalizing Q gives

$$\mathbb{E}\langle x, y \rangle^4 = 3\mathbb{E}(g^\top Q g)^2 = 3(\text{tr } Q)^2 + 6\text{tr } Q^2 \leq 9.$$

Homogeneity proves (2.8).

2. The matrix second moment.

For unit x , Cauchy–Schwarz in each summand gives

$$x^\top (\mathbb{E}Y^2) x = \sum_{a=1}^r \mathbb{E}[y_a^2 \langle x, y \rangle^2] \leq \sum_{a=1}^r (\mathbb{E}y_a^4)^{1/2} (\mathbb{E}\langle x, y \rangle^4)^{1/2} \leq 9r.$$

3. Center.

Finally $\mathbb{E}X^2 = \mathbb{E}Y^2 - 2\mathbb{E}Y + I_r = \mathbb{E}Y^2 - I_r$. □

The common-factor case $y = \eta\xi$ has $\|y\|_{L_p} = \|\eta\|_{L_p} \|\xi\|_{L_p}$, of order $\sqrt{rp} + p$. It calibrates the uniform order of Lemma 2.2, but is not used to restrict its upper-bound hypotheses.

3 Chain A: matrix moments, truncation, and concentration

Baseline A. The constants in this chain remain $9r$ and $72(r + s)$. The Wick-assisted concentration variant is separated in Appendix D. Chain A never constructs a graded multiplication operator. It converts the common row estimates into operator-valued moments of one rank-one matrix, truncates only for the exponential series, and uses matrix Laplace concentration. The final statement is for the original, unmodified sketch.

The overall architecture has direct precedents in the Khatri–Rao literature identified in Section 1.3. Reproving the concentration step below does not make that step, or the use of a truncated sample covariance, a new general method. The displayed constants belong to the implementation in this note.

3.1 A1. Keep matrix orientation when estimating powers

Lemma 3.1 (Explicit one-row matrix moments). *For $Y = yy^\top$ and every integer $p \geq 3$,*

$$\mathbb{E}Y^p \preceq 3[(8p - 11)r + (4p - 6)^2]^{p-1}I_r, \quad (3.1)$$

$$\mathbb{E}Y^p \preceq \frac{p!}{2} 9r [72(r + p)]^{p-2}I_r. \quad (3.2)$$

For $p = 2$, $\mathbb{E}Y^2 \preceq 9rI_r$.

Proof. **1. Bound each directional moment of the matrix power.**

The rank-one identity

$$Y^p = \|y\|^{2p-2}yy^\top$$

lets us keep the direction x until after expectation. For any unit x ,

$$\begin{aligned} x^\top(\mathbb{E}Y^p)x &= \mathbb{E}[\|y\|^{2p-2}\langle x, y \rangle^2] \\ &\leq (\mathbb{E}\|y\|^{4p-4})^{1/2}(\mathbb{E}\langle x, y \rangle^4)^{1/2} \\ &\leq 3\|y\|_{L_{4p-4}}^{2p-2} \\ &\leq 3[(8p - 11)r + (4p - 6)^2]^{p-1}. \end{aligned}$$

The second line is Cauchy–Schwarz; the third bounds the directional factor by Lemma 2.3; the last applies Lemma 2.2 at exponent $4p - 4$. Since x is arbitrary, this proves (3.1) as a Loewner bound.

2. Pass to the uniform moment scale $72(r + p)$.

The scalar inequality

$$3[(8p - 11)r + (4p - 6)^2]^{p-1} \leq \frac{p!}{2} 9r [72(r + p)]^{p-2} \quad (3.3)$$

is proved for every $r \geq 1, p \geq 3$ in Appendix B, giving (3.2). The $p = 2$ case is Lemma 2.3. $\square$

The distinction is between $\|\mathbb{E}Y^p\|$ and $\mathbb{E}\|Y\|^p$. Replacing Y by its norm before taking expectation discards the direction used above.

3.2 A2. Truncate without paying the full cutoff height

The untruncated Y need not have a finite positive exponential moment. For example, when $r = 1$ and $y = gh$, $Y = g^2h^2$ has infinite moment-generating function at every positive argument. Conditional on g , the Gaussian integral in h already diverges whenever $2\theta g^2 \geq 1$, an event of positive probability. Thus a positive-matrix exponential theorem cannot be applied to Y directly.

Lemma 3.2 (All-order moments of a bounded auxiliary matrix). *Fix an integer $s \geq 2$ and define*

$$R_s = 72(r + s), \quad L_s = sR_s, \quad W = Y\mathbf{1}_{\{\|y\|^2 \leq L_s\}}, \quad \Sigma = \mathbb{E}W. \quad (3.4)$$

Then

$$\mathbb{P}\{W \neq Y\} \leq e^{-2s}, \quad (3.5)$$

$$0 \preceq I_r - \Sigma \preceq 3e^{-s}I_r, \quad (3.6)$$

$$\mathbb{E}W^p \preceq \frac{p!}{2} 9r R_s^{p-2} I_r \quad \text{for every integer } p \geq 2. \quad (3.7)$$

Proof. **1. Bound the event on which the sample is changed.**

At exponent $2s$, Lemma 2.2 gives

$$\|\|y\|^2\|_{L_s} \leq r(4s - 3) + (2s - 2)^2 \leq 4s(r + s).$$

Markov's inequality therefore gives

$$\mathbb{P}\{\|y\|^2 > L_s\} \leq \left(\frac{4s(r + s)}{72s(r + s)} \right)^s = 18^{-s} \leq e^{-2s}.$$

2. Charge the covariance bias in each direction.

For a unit vector x , Lemma 2.3 and Cauchy–Schwarz imply

$$x^\top (I_r - \Sigma)x = \mathbb{E}[\langle x, y \rangle^2 \mathbf{1}_{\{\|y\|^2 > L_s\}}] \leq 3\mathbb{P}\{\|y\|^2 > L_s\}^{1/2} \leq 3e^{-s}.$$

The integrand is nonnegative, proving both sides of (3.6).

3. Prove the matrix moment bound through order s .

For $p = 2$, $\mathbb{E}W^2 \preceq \mathbb{E}Y^2 \preceq 9rI_r$. For $3 \leq p \leq s$, the scalar event in (3.4) gives $W^p \preceq Y^p$ pointwise. Lemma 3.1 and $p \leq s$ yield

$$\mathbb{E}W^p \preceq \frac{p!}{2} 9r [72(r + p)]^{p-2} I_r \preceq \frac{p!}{2} 9r R_s^{p-2} I_r.$$

4. Extend to every order required by the exponential series.

For an integer $p > s$, spectral calculus on $0 \preceq W \preceq L_s I_r$ gives

$$W^p \preceq L_s^{p-s} W^s.$$

Consequently,

$$\mathbb{E}W^p \preceq \frac{s!}{2} 9r R_s^{s-2} L_s^{p-s} I_r = \frac{s! s^{p-s}}{2} 9r R_s^{p-2} I_r \preceq \frac{p!}{2} 9r R_s^{p-2} I_r.$$

The last inequality follows by comparing each factor of $(s+1)(s+2) \cdots p$ with s . This proves all of (3.7), not just the first s moments. $\square$

The almost-sure cutoff height is $L_s = sR_s$, but the matrix moment-growth scale is only R_s . Charging L_s as the bounded-summand scale would lose the factor s .

3.3 A3. A matrix Laplace step with valid centering

The classical matrix-analysis input is Lieb’s trace-concavity theorem: for a fixed self-adjoint H , the map

$$Q \mapsto \operatorname{tr} \exp(H + \log Q) \quad (3.8)$$

is concave on the positive-definite cone [15, 16]. We use no assumption that a random matrix commutes with its expectation.

Lemma 3.3 (Matrix Laplace iteration with an offset). *For independent self-adjoint random matrices $A_1, \dots, A_m$ with finite matrix exponential moments, and deterministic self-adjoint H ,*

$$\mathbb{E} \operatorname{tr} \exp \left(H + \sum_{i=1}^m A_i \right) \leq \operatorname{tr} \exp \left(H + \sum_{i=1}^m \log \mathbb{E} e^{A_i} \right). \quad (3.9)$$

Proof. For one random matrix A , write $Q = e^A$. Jensen’s inequality applied to (3.8) gives

$$\mathbb{E} \operatorname{tr} \exp(H + A) \leq \operatorname{tr} \exp(H + \log \mathbb{E} e^A).$$

For the sum, condition first on $A_1, \dots, A_{m-1}$, treating all their terms as part of H . Independence makes $\log \mathbb{E} e^{A_m}$ deterministic. Repeat for $A_{m-1}, \dots, A_1$. This is the matrix Laplace iteration used by Tropp; no reordering of products or commuting-mean identity is involved [16]. $\square$

The next lemma is the positive-raw-moment version of the usual subexponential matrix Bernstein argument [16, Theorem 6.2 and Lemma 6.8]. Its proof uses the deterministic offset in Lemma 3.3 so that no commutation of a sample and its mean is assumed; it is not presented as a new abstract concentration inequality.

Lemma 3.4 (Concentration from positive matrix moments). *Let $W_1, \dots, W_m$ be independent, identically distributed bounded positive semidefinite $r \times r$ matrices with $\mathbb{E} W_i = \Sigma$. Suppose that for $p \geq 2$,*

$$\mathbb{E} W_i^p \preceq \frac{p!}{2} v R^{p-2} I_r.$$

Then for every $u > 0$,

$$\mathbb{P} \left\{ \left\| \frac{1}{m} \sum_i W_i - \Sigma \right\| > \sqrt{\frac{2vu}{m}} + \frac{2Ru}{m} \right\} \leq 2re^{-u}. \quad (3.10)$$

Proof. 1. Bound the logarithm of the uncentered exponential moment.

For $|\theta|R < 1$, positivity of each W_i^p and the exponential series imply

$$\mathbb{E} e^{\theta W_i} - I_r - \theta \Sigma \preceq \sum_{p \geq 2} \frac{|\theta|^p}{p!} \mathbb{E} W_i^p \preceq \frac{\theta^2 v}{2(1 - |\theta|R)} I_r.$$

For positive definite Q , spectral calculus gives $\log Q \preceq Q - I_r$. Therefore

$$\log \mathbb{E} e^{\theta W_i} \preceq \theta \Sigma + \frac{\theta^2 v}{2(1 - |\theta|R)} I_r. \quad (3.11)$$

2. Center by a deterministic offset inside the trace exponential.

In Lemma 3.3, take $A_i = \theta W_i$ and $H = -\theta m \Sigma$. Using (3.11),

$$\mathbb{E} \operatorname{tr} \exp \left(\theta \sum_i (W_i - \Sigma) \right) \leq r \exp \left(\frac{m \theta^2 v}{2(1 - |\theta|R)} \right).$$

Here a Loewner bound on the exponent gives a bound on each eigenvalue and hence on the trace exponential. No operator monotonicity of the exponential is needed.

3. Convert the exponential bound to the two spectral tails.

For a self-adjoint D and $\theta > 0$, the event $\lambda_{\max}(D) > x$ implies $\text{tr } e^{\theta D} > e^{\theta x}$. Markov, followed by $\theta = x/(mv + Rx)$, gives

$$\mathbb{P} \left\{ \lambda_{\max} \left(\sum_i (W_i - \Sigma) \right) > x \right\} \leq r \exp \left(-\frac{x^2}{2(mv + Rx)} \right).$$

Use negative θ for the lower spectral tail and add the probabilities. Finally $x = \sqrt{2m v u} + 2Ru$ satisfies $x^2 \geq 2u(mv + Rx)$, proving (3.10). $\square$

3.4 A4. Restore the original samples and solve for m

Theorem 3.5 (Explicit Chain A certificate). *Let $t = \log(4r/\delta)$ and $s = \lceil \log(4m/\delta) \rceil$. Then*

$$\mathbb{P} \left\{ \|Z\| > \sqrt{\frac{18rt}{m}} + \frac{144(r+s)t}{m} + 3e^{-s} \right\} \leq \delta. \quad (3.12)$$

In particular, having the displayed allowance at most ε is an explicit sufficient condition for the embedding guarantee.

Proof. The integer s is at least 2.

1. Replace every sample by its truncation.

Apply Lemma 3.2 separately to the independent samples, with the same deterministic cutoff. A union bound gives

$$\mathbb{P}\{\exists i : W_i \neq Y_i\} \leq m e^{-2s} \leq \frac{\delta^2}{16m} \leq \frac{\delta}{16}.$$

2. Concentrate the truncated average.

Lemma 3.4, with $v = 9r$, $R = R_s = 72(r+s)$, and $u = t$, fails with probability at most $2re^{-t} = \delta/2$.

3. Return to the original sketch.

On the intersection of the two good events, every W_i equals Y_i , so

$$\begin{aligned} \|Z\| &\leq \left\| \frac{1}{m} \sum_i W_i - \Sigma \right\| + \|\Sigma - I_r\| \\ &\leq \sqrt{\frac{18rt}{m}} + \frac{144(r+s)t}{m} + 3e^{-s}. \end{aligned}$$

The total failure probability is less than δ . The conclusion concerns the original S , not an estimator that clips its rows during execution. $\square$

Proposition 3.6 (The common sample order follows from Chain A). *The explicit certificate (3.12) implies (1.2) with a universal constant.*

Proof. **1. Simplify the allowance to two monotone terms.**

Since $s \leq \log(4m/\delta) + 1$ and $3e^{-s} \leq 3\delta/(4m)$, the allowance in (3.12) is at most

$$C_1 \left[\sqrt{\frac{rt}{m}} + \frac{(r + \log(4m/\delta) + 1)t}{m} \right] \quad (3.13)$$

for an absolute C_1 . Both terms decrease with real $m \geq 1$; for the second, differentiate $(r + \log(4m/\delta) + 1)/m$.

2. Evaluate at the candidate sample size.

Set

$$m_0 = K \left(\frac{rt}{\varepsilon^2} + \frac{t^2}{\varepsilon} \right) = K \varepsilon^{-2} t(r + \varepsilon t), \quad K \geq 1.$$

Here $t > 1$, $r \leq e^t$, and $t \leq e^t$, so

$$\log(4m_0/\delta) + 1 \leq \log K + 2 \log(1/\varepsilon) + 3t + \log 2 + 1.$$

3. Bound each term and fix the constant.

First,

$$\sqrt{\frac{rt}{m_0}} \leq \frac{\varepsilon}{\sqrt{K}}.$$

Second,

$$\begin{aligned} \frac{(r + \log(4m_0/\delta) + 1)t}{m_0} &\leq \frac{\varepsilon^2}{K} \frac{r + 3t + 2 \log(1/\varepsilon) + \log K + \log 2 + 1}{r + \varepsilon t} \\ &\leq \frac{C_2(1 + \log K)}{K} \varepsilon. \end{aligned}$$

For the last step, use $r \geq 1$, $r + \varepsilon t \geq \varepsilon t$, and $\varepsilon \log(1/\varepsilon) \leq 1/e$. Choose K so that $C_1[K^{-1/2} + C_2(1 + \log K)/K] \leq 1$. Monotonicity of (3.13) handles every larger integer m . $\square$

4 Chain B: exact graded multiplication and vacuum propagation

Baseline B. Retain $v_r = 9r - 1$ and the original $B_q(r; \lambda)$. The Wick–Hermite refinements belong to the separate Chain C. Chain B uses the four core stages of the framework: exact multiplication, reachable graded states, coefficient-sensitive contractions, and vacuum-moment propagation. No row is clipped and no matrix Laplace theorem is invoked. The central estimate charges each already-active sample its own local degree *before* summing over samples.

4.1 B1. The original-law multiplication identity

Multiplication in L_2 and orthogonal-polynomial coordinates are classical; Wiener [17] is a source for homogeneous Gaussian chaos, and Accardi–Bożejko [18] provide a probability-law/creation-operator representation precedent. We use these as background attributions, not as claims that either source proves the present embedding theorem. The identity below is simply Parseval applied to multiplication. Let

$$\mathcal{H} = \mathbb{R}^r \otimes \bigotimes_{i=1}^m L_2(\gamma_{n_1} \otimes \gamma_{n_2}), \quad \Omega = 1.$$

The external factor $\mathbb{R}^r$ carries the matrix indices, one L_2 factor carries the randomness of one sample, and the vacuum Ω is the constant function 1 in every sample register. Define $\mathcal{M}_Z f = Zf$ on the vector polynomials in this Hilbert space. For an external standard basis vector e_a ,

$$\mathcal{M}_Z^j(e_a \otimes \Omega)(g_1, h_1, \dots, g_m, h_m) = Z^j e_a.$$

All of these polynomials are square integrable. The pointwise matrix Z is symmetric because each $Y_i = y_i y_i^\top$ is symmetric. Thus $\mathcal{M}_Z$ is symmetric on its invariant polynomial domain; we do not assert self-adjointness of this unbounded restriction. The identity follows directly from $e_a^\top Z^{2q} e_a = \|Z^q e_a\|^2$ pointwise:

$$\mathbb{E} e_a^\top Z^{2q} e_a = \langle e_a \otimes \Omega, \mathcal{M}_Z^{2q}(e_a \otimes \Omega) \rangle_{\mathcal{H}} = \|\mathcal{M}_Z^q(e_a \otimes \Omega)\|_{\mathcal{H}}^2.$$

Summing over a gives the identity the chain rests on:

$$\mathbb{E} \operatorname{tr} Z^{2q} = \sum_{a=1}^r \|\mathcal{M}_Z^q(e_a \otimes \Omega)\|_{\mathcal{H}}^2. \quad (4.1)$$

The factor r that will eventually arise counts external vacuum seeds. We do not take a trace over the generally much larger Hermite occupation space. Every visit to sample i still uses the same g_i, h_i .

4.2 B2. Hermite degree, active samples, and the exact cutoff

Let $h_\ell = \operatorname{He}_\ell / \sqrt{\ell!}$ denote the normalized probabilists' Hermite polynomial. Its recurrence is

$$x h_\ell(x) = \sqrt{\ell+1} h_{\ell+1}(x) + \sqrt{\ell} h_{\ell-1}(x), \quad h_{-1} = 0. \quad (4.2)$$

Products of these functions form an orthonormal basis in each Gaussian sample space. Let N_i record the total local degree in (g_i, h_i) . Let P_i be the orthogonal projection onto the local constant function, tensored with identities on all other factors, and put $Q_i = I - P_i$.

Definition 4.1 (An active sample). A sample is active in a product Hermite component when its local factor is nonconstant, equivalently when it lies in $Q_i L_2$. The exact active set is the set of such indices. Its size is a second grading, distinct from total degree.

An active set is not a history of labels that have ever occurred. A sample can return to its local vacuum, and then be activated again by a later multiplication.

There are exact sign-flip symmetries. Replacing the entire vector g_i by $-g_i$ changes y_i to $-y_i$ but leaves Y_i , X_i , and Z unchanged; the same holds for h_i . We restrict to functions having even total Hermite degree separately in every g_i and every h_i register. The vacuum belongs to this subspace, and multiplication by Z preserves it.

After j multiplications from vacuum,

$$\sum_i N_i \leq 4j, \quad \#\{\text{active samples}\} \leq j. \quad (4.3)$$

Indeed, X_i has polynomial degree at most four and acts on only one sample.

Let Π_q project onto the symmetry subspace with total degree at most $4q$ and at most q active samples, and set

$$M_q = \Pi_q \mathcal{M}_Z \Pi_q. \quad (4.4)$$

This finite-dimensional compression is symmetric, hence self-adjoint. Inducting on j using (4.3) gives

$$M_q^j(e_a \otimes \Omega) = \mathcal{M}_Z^j(e_a \otimes \Omega), \quad 0 \leq j \leq q. \quad (4.5)$$

Thus (4.1) remains exact with M_q in place of $\mathcal{M}_Z$. This is the elementary finite-moment compression principle as organized in the framework [1]; its proof here is the reachability induction just given. Neither that induction nor the underlying polynomial representation is claimed as a new general theorem. No bound on the full unbounded Gaussian multiplication operator is required.

4.3 B3. Split activation, return, and repeated visits

Write $\mathcal{X}_i$ for multiplication by X_i . Centering is the exact identity $P_i \mathcal{X}_i P_i = 0$. Therefore

$$\mathcal{X}_i = Q_i \mathcal{X}_i P_i + P_i \mathcal{X}_i Q_i + Q_i \mathcal{X}_i Q_i.$$

Define, on the common cutoff,

$$C = \Pi_q \sum_i Q_i \mathcal{X}_i P_i \Pi_q, \quad \mathcal{R} = \Pi_q \sum_i Q_i \mathcal{X}_i Q_i \Pi_q. \quad (4.6)$$

Then

$$M_q = \frac{1}{m}(C + C^* + \mathcal{R}). \quad (4.7)$$

The operator C increases active-sample count by one; C^* decreases it by one; $\mathcal{R}$ preserves the exact active set. These are actual full-space compressions, so using adjointness is legitimate. Let $\mathcal{K}_k$ denote the cutoff sector with exactly k active samples.

4.3.1 Activation and return are controlled by matrix variance

Lemma 4.2 (Fresh-sample contraction). *With $v_r = 9r - 1$, for $0 \leq k < \min(q, m)$,*

$$\|C : \mathcal{K}_k \rightarrow \mathcal{K}_{k+1}\| \leq \sqrt{(k+1)(m-k)v_r}. \quad (4.8)$$

The reverse map C^ has the same bound.*

Proof. 1. Compute the single-insertion Gram before the output cutoff.

For a sample currently in its local vacuum,

$$\begin{aligned} (Q_i \mathcal{X}_i P_i)^* (Q_i \mathcal{X}_i P_i) &= P_i \mathcal{X}_i Q_i \mathcal{X}_i P_i \\ &= P_i \mathcal{X}_i^2 P_i = (\mathbb{E} X_i^2) P_i \preceq v_r P_i. \end{aligned}$$

The middle equality uses $P_i \mathcal{X}_i P_i = 0$. It remains valid with arbitrary Hilbert-valued coefficients in the other sample registers. Output projection by Π_q can only decrease a norm.

2. Count the inputs that can meet at one output set.

Decompose $f = \sum_{|S|=k} f_S$ by exact active set; these inputs are orthogonal. For an output set T of size $k+1$, the contributing inputs are $f_{T \setminus \{i\}}$, one for each $i \in T$. Cauchy–Schwarz gives

$$\left\| \sum_{i \in T} \Pi_q Q_i \mathcal{X}_i P_i f_{T \setminus \{i\}} \right\|^2 \leq (k+1) v_r \sum_{i \in T} \|f_{T \setminus \{i\}}\|^2.$$

Sum over T . Each input set S appears once for each of the $m-k$ absent labels, yielding

$$\|Cf\|^2 \leq (k+1)(m-k) v_r \sum_{|S|=k} \|f_S\|^2.$$

Take square roots and then use adjointness. □

4.3.2 Repeated visits are controlled by local degree

For one sample define the row multiplier

$$Tf = y^\top f, \quad T^*T = \mathcal{M}_Y$$

as a quadratic-form identity on polynomials. All following estimates allow a Hilbert-valued coefficient space for the other registers. Hölder is applied only in the current sample variables, with that coefficient space kept intact. Before the finite cutoff, identities involving multipliers and their adjoints are understood as polynomial quadratic-form identities; after compression the adjoints are ordinary finite-dimensional ones.

Lemma 4.3 (Local degree-dependent row contraction). *Let f_d have homogeneous local Hermite degree $d \geq 1$. For every real $\lambda > 0$,*

$$\|Tf_d\|_2^2 \leq e^{2/\lambda} [(4\lambda d + 1)r + 4\lambda^2 d^2] \|f_d\|_2^2. \quad (4.9)$$

For a finite sum $f = \sum_{d \geq 1} f_d$ with no local vacuum component,

$$\|Tf\|_2^2 \leq 3e^{2/\lambda} \sum_{d \geq 1} [(4\lambda d + 1)r + 4\lambda^2 d^2] \|f_d\|_2^2. \quad (4.10)$$

Proof. 1. Choose the Hölder exponent from the actual local degree.

Set $p = \lambda d + 1 > 1$. Hölder and (2.2) give

$$\begin{aligned} \|y^\top f_d\|_2 &\leq \| \|y\| \|f_d\|_{L_{2p/(p-1)}} \|f_d\|_2 \\ &\leq \| \|y\| \|f_d\|_{L_{2p}} \left(1 + \frac{2}{\lambda d}\right)^{d/2} \|f_d\|_2. \end{aligned}$$

Lemma 2.2 bounds the square of the first factor by

$$r(4p-3) + (2p-2)^2 = (4\lambda d + 1)r + 4\lambda^2 d^2.$$

Since $(1 + 2/(\lambda d))^d \leq e^{2/\lambda}$, this proves (4.9). The multiplier and the polynomial use the same sample; no independence between y and f_d is assumed.

2. Sum homogeneous inputs using their output-degree overlap.

Every coordinate of y is a pure second-chaos polynomial. In fact it is a linear combination of $g_a h_b$, with distinct factor coordinates. Write $\mathcal{F}_e$ for the closed span of the components of exact local Hermite degree e , with arbitrary coefficients in the other registers; set $\mathcal{F}_e = \{0\}$ for $e < 0$. Applying the Hermite recurrence (4.2) twice shows

$$Tf_d \in \mathcal{F}_{d-2} \oplus \mathcal{F}_d \oplus \mathcal{F}_{d+2}.$$

At any fixed output degree, at most three input degrees contribute. Cauchy–Schwarz on those contributions, followed by orthogonality of the output degrees, gives $\|Tf\|_2^2 \leq 3 \sum_d \|Tf_d\|_2^2$. Now use (4.9). $\square$

Define the deterministic allowance

$$B_q(r; \lambda) = 3e^{2/\lambda} [(16\lambda + 1)rq + 64\lambda^2 q^2]. \quad (4.11)$$

The positive real λ is a proof parameter, not a property of the subspace. In particular,

$$B_q(r; 2) = 99erq + 768eq^2, \quad B_q(r; 1) = 51e^2rq + 192e^2q^2. \quad (4.12)$$

Lemma 4.4 (Sum of already-active sample interactions). *On the entire cutoff of (4.4),*

$$\|\mathcal{R}\| \leq B_q(r; \lambda). \quad (4.13)$$

Its restriction to the vacuum sector is zero.

Proof. 1. Fix the exact active set and expose the positive part.

On a support sector S with $k = |S| \leq q$, the quadratic form is

$$\langle f, \mathcal{R}f \rangle = \sum_{i \in S} \|T_i f\|_2^2 - k\|f\|_2^2. \quad (4.14)$$

This follows because $Q_i f = f$ for $i \in S$ and $Q_i f = 0$ otherwise, together with the one-sample identity $\langle f, \mathcal{X}_i f \rangle = \langle f, (\mathcal{M}_{Y_i} - I)f \rangle = \|T_i f\|_2^2 - \|f\|_2^2$. Thus $\mathcal{R} \succeq -kI$ on this sector.

2. Sum each label's actual degree before relaxing any budget.

Let $f = \sum_{\mathbf{d}} f_{\mathbf{d}}$ be the joint local-degree decomposition. The number operators N_i commute, and their joint eigenspaces are orthogonal. Apply (4.10) to each $i \in S$ and then combine the sums:

$$\begin{aligned} \sum_{i \in S} \|T_i f\|_2^2 &\leq 3e^{2/\lambda} \sum_{\mathbf{d}} \left[r \left(4\lambda \sum_{i \in S} d_i + k \right) + 4\lambda^2 \sum_{i \in S} d_i^2 \right] \|f_{\mathbf{d}}\|_2^2 \\ &\leq 3e^{2/\lambda} [(16\lambda + 1)rq + 64\lambda^2 q^2] \|f\|_2^2. \end{aligned}$$

The last inequality uses precisely

$$\sum_i d_i \leq 4q, \quad k \leq q, \quad \sum_i d_i^2 \leq \left(\sum_i d_i \right)^2 \leq 16q^2.$$

Charging every active label degree $4q$ separately would be an unnecessary extra factor of q ; the displayed summation avoids that loss.

3. Control both sides of the centered form.

Equation (4.14) and the last estimate give $-kI \preceq \mathcal{R} \preceq B_q I$. Since $B_q \geq q \geq k$, its operator norm is at most B_q . Compression by the cutoff preserves the form inequalities. Finally sum over orthogonal exact-support sectors. On the vacuum sector $S = \emptyset$, the operator vanishes. $\square$

4.4 B4. Propagate the vacuum rather than the worst state

Let $K = \min(q, m)$ and define the nonnegative symmetric tridiagonal matrix G_q on indices $0, \dots, K$ by

$$\begin{aligned} (G_q)_{00} &= 0, & (G_q)_{kk} &= B_q(r; \lambda)/m \quad (k \geq 1), \\ (G_q)_{k+1,k} &= (G_q)_{k,k+1} = \frac{\sqrt{(k+1)(m-k)}v_r}{m} \quad (0 \leq k < K). \end{aligned} \quad (4.15)$$

If $n(f) = (\|P_{\mathcal{K}_k} f\|)_{k=0}^K$ is the vector of sector norms, Lemmas 4.2 and 4.4 give

$$n(M_q f) \leq G_q n(f) \quad (4.16)$$

componentwise. Since G_q is entrywise nonnegative, iteration from $n(e_a \otimes \Omega) = e_0$ yields

$$\boxed{\mathbb{E} \operatorname{tr} Z^{2q} \leq r \|G_q^q e_0\|_2^2 = r (G_q^{2q})_{00}.} \quad (4.17)$$

This is the nonnegative vacuum-majorant rule used in the framework [1]. The comparison itself follows from block norms and positivity; the local activation and repeated-visit estimates supply its application-specific entries. All occurrences of e_0 in scalar majorants refer to the active-count vacuum; it is distinct from the external basis vectors e_a .

4.4.1 A closed-form consequence

The scalar Jacobi comparison uses the classical relation between orthogonal polynomials, finite Jacobi matrices, and Gaussian quadrature; see Golub–Welsch [19]. The vacuum weights, rather than the unweighted trace of that Jacobi matrix, are essential. The required finite-moment identity is proved explicitly below. Let

$$c_q = ((2q-1)!!)^{1/(2q)}, \quad \eta_q = (r/\delta)^{1/(2q)}. \quad (4.18)$$

Theorem 4.5 (Untruncated closed-form moment bound). *For every integer $q \geq 1$ and every $\lambda > 0$,*

$$\boxed{(\mathbb{E} \operatorname{tr} Z^{2q})^{1/(2q)} \leq r^{1/(2q)} \left[c_q \sqrt{\frac{v_r}{m}} + \frac{B_q(r; \lambda)}{m} \right].} \quad (4.19)$$

Consequently

$$\mathbb{P} \left\{ \|Z\| > \eta_q \left[c_q \sqrt{\frac{v_r}{m}} + \frac{B_q(r; \lambda)}{m} \right] \right\} \leq \delta. \quad (4.20)$$

Proof. **1. Dominate the sector matrix entrywise.**

Let J_q be the $(q+1) \times (q+1)$ Hermite Jacobi matrix with zero diagonal and $(J_q)_{k+1,k} = (J_q)_{k,k+1} = \sqrt{k+1}$. After padding G_q by zeros if $m < q$, it is entrywise at most

$$bI + aJ_q, \quad a = \sqrt{v_r/m}, \quad b = B_q(r; \lambda)/m,$$

because $(k+1)(m-k) \leq (k+1)m$ and $B_q \geq 0$ on the diagonal. The entries are nonnegative, so matrix powers preserve this comparison, and in particular $\|G_q^q e_0\| \leq \|(bI + aJ_q)^q e_0\|$.

2. Identify the comparison vacuum moment as a scalar Gaussian moment.

By (4.2), J_q is multiplication by a scalar standard Gaussian ξ compressed to Hermite degrees $0, \dots, q$. Each application of $b + a\xi$ raises the degree by at most one, so q applications starting from $h_0 = 1$ never leave degrees $0, \dots, q$, and the compression changes nothing. Thus

$$\|(bI + aJ_q)^q e_0\|^2 = \mathbb{E}(b + a\xi)^{2q}.$$

3. Separate the two rates and apply Markov.

Minkowski in scalar L_{2q} gives

$$\|G_q^q e_0\|^{1/q} \leq b + a\|\xi\|_{L_{2q}} = b + ac_q.$$

Combine this with (4.17) to obtain (4.19). Finally, $\mathbb{P}\{\|Z\| > x\} \leq x^{-2q} \mathbb{E} \operatorname{tr} Z^{2q}$ at the displayed threshold proves the tail bound (4.20), since $\eta_q^{2q} = r/\delta$. $\square$

The scalar Gaussian ξ describes only the comparison Jacobi operator. It does not replace any original tensor factor or any repeated sample in the original law.

Corollary 4.6 (The common sample order follows from Chain B). *The framework proof implies (1.2). An explicit sufficient row count for any chosen $q \geq 1, \lambda > 0$ is*

$$m \geq \left\lceil \left(\frac{a_q + \sqrt{a_q^2 + 4\varepsilon b_q}}{2\varepsilon} \right)^2 \right\rceil, \quad a_q = \eta_q c_q \sqrt{v_r}, \quad b_q = \eta_q B_q(r; \lambda). \quad (4.21)$$

Proof. 1. The exact threshold.

Equation (4.21) solves $a_q/\sqrt{m} + b_q/m \leq \varepsilon$ exactly as a quadratic inequality in $\sqrt{m}$.

2. The sample-size order.

Take $q = \lceil \log(r/\delta) \rceil \geq 1$ and $\lambda = 2$. Then $\eta_q \leq \sqrt{e}$, $c_q \leq \sqrt{2q}$, and $q \leq 2t$ for $t = \log(4r/\delta) > 1$. The allowance is therefore bounded by an absolute constant times

$$\sqrt{rq/m} + rq/m + q^2/m.$$

Making each term at most a fixed fraction of ε requires the sample order $\varepsilon^{-2}rt + \varepsilon^{-1}rt + \varepsilon^{-1}t^2$. Since $\varepsilon < 1$, the middle term is absorbed by the first, which gives (1.2). $\square$

4.5 A stronger certificate: use the budget at each stage

The fixed cutoff at degree $4q$ is exact, but its largest repeated-visit cost need not be charged at every earlier step. A prefix of length $j - 1$ only needs the next cutoff at degree $4j$ and at most j active samples.

Fix the terminal q and $K = \min(q, m)$. Define a symmetric nonnegative matrix $G^{[j]}$ on $0, \dots, K$ for $1 \leq j \leq q$ by

$$\begin{aligned} (G^{[j]})_{00} &= 0, & (G^{[j]})_{kk} &= B_j(r; \lambda)/m \quad (k \geq 1), \\ (G^{[j]})_{k+1,k} &= (G^{[j]})_{k,k+1} = \frac{\sqrt{(k+1)(m-k)v_r}}{m}. \end{aligned} \quad (4.22)$$

Entries beyond currently reachable sectors are harmless padding. Propagate in the stated order:

$$w^{(0)} = e_0, \quad w^{(j)} = G^{[j]} w^{(j-1)}. \quad (4.23)$$

Theorem 4.7 (Stage-dependent vacuum certificate). *For the original independent Gaussian law,*

$$\boxed{\mathbb{E} \operatorname{tr} Z^{2q} \leq r \|G^{[q]} G^{[q-1]} \dots G^{[1]} e_0\|_2^2.} \quad (4.24)$$

In particular, the computable inequality

$$r \|G^{[q]} \dots G^{[1]} e_0\|_2^2 \leq \delta \varepsilon^{2q} \quad (4.25)$$

is sufficient for $\mathbb{P}\{\|Z\| > \varepsilon\} \leq \delta$. At fixed q, λ, r, m , this certificate is no worse than the stationary certificate (4.17).

Proof. 1. One exact stage.

For one external seed, put $f_j = \mathcal{M}_Z^j(e_a \otimes \Omega)$. By (4.3), f_{j-1} belongs to the cutoff with degree $4(j-1)$ and active count at most $j-1$, while f_j lies in the next cutoff. Let $M_j = \Pi_j \mathcal{M}_Z \Pi_j$ use that next cutoff. Then $f_j = M_j f_{j-1}$ exactly.

2. Propagate sector norms in the stated order.

Apply Lemmas 4.2 and 4.4 with parameter j :

$$n(f_j) \leq G^{[j]} n(f_{j-1}).$$

Induction gives $n(f_q) \leq w^{(q)}$ componentwise. Sum squared norms over all r external seeds and use (4.1) to obtain (4.24). Markov then yields (4.25).

3. Compare with the stationary certificate.

Finally $B_j(r; \lambda) \leq B_q(r; \lambda)$ implies $G^{[j]} \leq G_q$ entrywise. Positivity and induction give $w^{(q)} \leq G_q^q e_0$ componentwise. $\square$

Remark 4.8 (Order and energy are essential). The matrices $G^{[j]}$ generally do not commute. The terminal quantity is the squared norm of an *ordered product*; it is not a product of top eigenvalues, a power of an average, or an unordered product of return costs.

To express the argument as an energy, define for a nonnegative sector vector z

$$\mathcal{E}_j(z) = \|G^{[q]} \dots G^{[j+1]} z\|_2^2, \quad \mathcal{E}_q(z) = \|z\|^2.$$

The componentwise step inequality and nonnegativity give $\mathcal{E}_j(n(f_j)) \leq \mathcal{E}_{j-1}(n(f_{j-1}))$. This is the framework's weighted-energy viewpoint, with a specified finite certificate rather than a formal re-expression of the unknown exact moment.

4.6 An exact limited-independence consequence

Corollary 4.9 (Moment matching of sample blocks). *Fix q . Suppose each block (g_i, h_i) has the correct two-factor Gaussian marginal law, and any set of at most $2q$ distinct blocks has the same joint law as under full independence. Then $\mathbb{E} \operatorname{tr} Z^{2q}$ is unchanged. All three framework certificates at this q therefore remain valid.*

Proof. Expand $\operatorname{tr}(\sum_i X_i)^{2q}$. Each word uses at most $2q$ distinct block labels. Its expectation is fixed by their joint law, even if a label is repeated many times within the word. Sum the finitely many equal expectations and divide by m^{2q} . The operator proof is carried out on the independent reference law; only its resulting moment is transferred. $\square$

This does not provide a finite-seed Gaussian generator or transfer the lower bound. No concentration or cutoff argument on the dependent law is asserted.

5 Chain C: the combined Wick–Hermite framework

A third proof chain, not a replacement for B

Wick contractions determine sharper local radial and covariance bounds. Those bounds enter B’s original-law Hermite multiplication, active-register cutoff, and ordered vacuum propagation. This hybrid never clips an outcome and never uses matrix Laplace concentration. Its constants carry a separate superscript (*C*); the constants and proofs of A and B remain unchanged.

The starting model is still (1.1). The coefficient matrices B_a are still the arbitrary Frobenius-orthonormal matrices of (2.4). No common Schmidt basis is assumed in an upper bound. Wick here means joint Gaussian product-moment contraction [14, 21], not replacement of ordinary multiplication by the Wick product with its contraction terms removed. The radialization idea is recorded in [22]; the particular Khatri–Rao refinement is supplied by [23]. Both are provenance, not claims of external validation.

5.1 C1. Contract jointly and retain the sharp radial envelope

For a positive integer p , define

$$\kappa_p = (2p - 1)!!, \quad D_{r,p} = \prod_{\ell=0}^{p-1} (r + 2\ell), \quad M_p(r) = (\kappa_p D_{r,p})^{1/p}. \quad (5.1)$$

Here κ_p counts scalar pairings. It is not the moment root c_q that appears later in the scalar Jacobi comparison. Set $\kappa_0 = D_{r,0} = 1$.

Theorem 5.1 (Sharp uniform radial moments). *For every integer $p \geq 1$,*

$$\mathbb{E} \|y\|^{2p} \leq \kappa_p D_{r,p}. \quad (5.2)$$

For every r , equality holds on the common-factor subspace $Ue_a = e_a \otimes e_1$. Moreover,

$$\|y\|_{L_{2p}}^2 \leq M_p(r) \leq p(r + p - 1). \quad (5.3)$$

Proof. 1. Contract one scalar projection exactly.

For $x \in \mathbb{R}^r$, put $B_x = \sum_a x_a B_a$. Then $\|B_x\|_F^2 = \|x\|^2$. Conditional Gaussian integration, equivalently Wick’s rule, gives

$$\mathbb{E}(g^\top B_x h)^{2p} = \kappa_p \mathbb{E} \left(\sum_j \lambda_j \gamma_j^2 \right)^p,$$

where $\lambda_j \geq 0$ are the squared singular values of B_x , their sum is $\|x\|^2$, and the γ_j are independent standard Gaussians. For $x \neq 0$, write $w_j = \lambda_j / \|x\|^2$, so $\sum_j w_j = 1$. Convexity gives

$$\left(\sum_j w_j \gamma_j^2 \right)^p \leq \sum_j w_j \gamma_j^{2p}.$$

Taking expectations (and handling $x = 0$ separately) yields

$$\mathbb{E} \langle x, y \rangle^{2p} \leq \kappa_p^2 \|x\|^{2p}. \quad (5.4)$$

For rank-one B_x the inequality is an equality. All repeated Gaussian occurrences were integrated jointly.

2. Recover the radial moment without a coordinatewise norm bound.

Let $\zeta \sim N(0, I_r)$ be independent of y . Conditional on y ,

$$\mathbb{E}_\zeta \langle \zeta, y \rangle^{2p} = \kappa_p \|y\|^{2p}.$$

Apply (5.4) conditional on ζ , then integrate:

$$\kappa_p \mathbb{E} \|y\|^{2p} \leq \kappa_p^2 \mathbb{E} \|\zeta\|^{2p} = \kappa_p^2 D_{r,p}.$$

The chi-square moment $D_{r,p}$ follows either from Wick pairing counts or Gaussian polar integration. Divide by κ_p . This auxiliary-Gaussian argument is equivalent to the spherical averaging in [22, projected radial moments]; ζ is a proof device, not an additional sketch factor.

3. Check equality and the elementary envelope.

For $y = \eta\xi$, with independent $\eta \sim N(0, 1)$ and $\xi \sim N(0, I_r)$, both factors in the radial moment separate, giving $\kappa_p D_{r,p}$ exactly. Finally the arithmetic means of the factors in κ_p and $D_{r,p}$ are p and $r + p - 1$. AM–GM proves (5.3). $\square$

Remark 5.2 (Why this is an improvement in constants, not a new rate). For r much larger than p , the exact envelope is asymptotically $\kappa_p^{1/p} r$, with $\kappa_p^{1/p} \sim 2p/e$. The preceding baseline is approximately $4pr$. The local leading ratio therefore approaches $2e$ when both approximations apply. The established order $\sqrt{rp} + p$ has not changed. The exact product also retains the regime in which p is comparable to or larger than r .

5.2 C2. Keep orthogonal mixed contractions

A bound on each directional fourth moment alone gives the earlier $9r$. To improve it, keep the orthogonality between distinct subspace coefficient matrices while doing the contractions.

For real matrices A, B of the same shape, define

$$Q(A, B) = \text{tr}(A^\top AB^\top B) + \text{tr}(AA^\top BB^\top) + \text{tr}((A^\top B)^2). \quad (5.5)$$

The last summand can have either sign. Wick–Isserlis gives the exact identity

$$\mathbb{E}(g^\top Ah)^2 (g^\top Bh)^2 = \|A\|_F^2 \|B\|_F^2 + 2\langle A, B \rangle_F^2 + 2Q(A, B). \quad (5.6)$$

One direct verification conditions on h . The Gaussian moment in g is $\|Ah\|^2 \|Bh\|^2 + 2(h^\top A^\top Bh)^2$; expanding its two quadratic-form moments in h gives (5.6).

Lemma 5.3 (Orthogonal mixed fourth moment). *If $\|A\|_F = \|B\|_F = 1$ and $\langle A, B \rangle_F = 0$, then*

$$\boxed{\mathbb{E}(g^\top Ah)^2 (g^\top Bh)^2 \leq 4.} \quad (5.7)$$

The constant 4 in this two-direction statement is attained for suitable 2×2 coefficient matrices.

Proof. 1. Use singular coordinates for A .

By orthogonal invariance, A is rectangular diagonal with singular values $s_i \geq 0$, $\sum_i s_i^2 = 1$. Write b_{ij} for the entries of B in these bases. On each off-diagonal pair (b_{ij}, b_{ji}) inside the diagonal support, the quadratic form Q has matrix

$$\begin{pmatrix} s_i^2 + s_j^2 & s_i s_j \\ s_i s_j & s_i^2 + s_j^2 \end{pmatrix}.$$

Its top eigenvalue is $s_i^2 + s_j^2 + s_i s_j \leq \frac{3}{2}(s_i^2 + s_j^2) \leq \frac{3}{2}$. Entries outside such a pair have coefficient at most 1 (or zero).

2. Use coefficient orthogonality on the diagonal.

The diagonal contribution is $3 \sum_i s_i^2 b_{ii}^2$, with $\sum_i s_i b_{ii} = 0$. For $u_i = s_i b_{ii}$, its positive and negative parts have equal total mass. Cauchy–Schwarz gives $\sum_i |u_i| \leq (\sum_i b_{ii}^2)^{1/2}$. Therefore

$$\sum_i u_i^2 \leq \left(\sum_{u_i > 0} u_i \right)^2 + \left(\sum_{u_i < 0} |u_i| \right)^2 \leq \frac{1}{2} \sum_i b_{ii}^2.$$

The diagonal contribution is at most $\frac{3}{2} \sum_i b_{ii}^2$. Combining all entries gives $Q(A, B) \leq \frac{3}{2} \|B\|_F^2 = \frac{3}{2}$.

3. Finish and check sharpness.

Equation (5.6) becomes $1 + 2Q \leq 4$. Equality holds for $A = 2^{-1/2} \text{diag}(1, 1)$ and $B = 2^{-1/2} \text{diag}(1, -1)$. $\square$

Corollary 5.4 (Uniform matrix second moments). *With*

$$\nu_r^{(C)} = 4r + 5, \quad v_r^{(C)} = 4r + 4, \quad (5.8)$$

one has

$$\boxed{\mathbb{E}Y^2 \preceq \nu_r^{(C)} I_r, \quad \mathbb{E}X^2 \preceq v_r^{(C)} I_r.} \quad (5.9)$$

Proof. Fix a unit x and rotate the external coefficient basis so that its first vector is x . The corresponding first coefficient matrix is B_x , and the other $r - 1$ coefficient matrices are Frobenius-orthogonal to it. Its own fourth moment is at most 9, by (5.4) with $p = 2$; each mixed fourth moment is at most 4 by Lemma 5.3. Hence

$$x^\top (\mathbb{E}Y^2) x = \mathbb{E} \|y\|^2 \langle x, y \rangle^2 \leq 9 + 4(r - 1) = 4r + 5.$$

Finally $\mathbb{E}X^2 = \mathbb{E}(Y - I_r)^2 = \mathbb{E}Y^2 - I_r$. $\square$

5.2.1 Optional exact covariance information

The Wick calculation also provides exact directional covariance data. Set

$$F_g = \sum_a B_a B_a^\top, \quad F_h = \sum_a B_a^\top B_a.$$

For unit x ,

$$\begin{aligned} x^\top (\mathbb{E}Y^2) x &= r + 2 + 2 \left[\text{tr}(B_x^\top B_x F_h) + \text{tr}(B_x B_x^\top F_g) \right. \\ &\quad \left. + \sum_a \text{tr}((B_x^\top B_a)^2) \right]. \end{aligned} \quad (5.10)$$

It follows by summing (5.6); $\sum_a \langle B_x, B_a \rangle_F^2 = 1$. An exactly computed or certified upper bound on $\|\mathbb{E}X^2\|$ can replace $v_r^{(C)}$ in the activation bound below. This does not automatically replace $\nu_r^{(C)}$ in every higher-moment estimate of Chain A.

Remark 5.5 (Radial sharpness does not identify the worst covariance). For the common-factor family $y = \eta \xi$, one has $\mathbb{E}Y^2 = 3(r + 2)I_r$ and $\mathbb{E}X^2 = (3r + 5)I_r$. That does not justify the universal allowance $3r + 5$. For example, let $r = 2$ and

$$B_1 = \text{diag}(a, b), \quad B_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad a^2 + b^2 = 1, \quad ab = \frac{1}{12}.$$

These two matrices are Frobenius-orthonormal. Wick gives $\mathbb{E}y_1^4 = 107/12$, $\mathbb{E}y_1^2 y_2^2 = 19/6$, and hence $e_1^\top (\mathbb{E}X^2) e_1 = 133/12 > 11 = 3r + 5$. The bound $4r + 4$ is a valid envelope, not a claimed sharp covariance constant.

5.3 C3. Insert the Wick bounds into the Hermite framework

The probability space, multiplication operator, sign-flip symmetries, cutoff Π_q , and compressed matrix M_q are exactly those defined and proved in Section 4. In particular,

$$\mathbb{E} \operatorname{tr} Z^{2q} = \sum_{a=1}^r \|M_q^q(e_a \otimes \Omega)\|^2, \quad mM_q = C + C^* + \mathcal{R}.$$

The letter C inside this decomposition denotes the activation operator of (4.6), not the proof-chain label. The operator itself is unchanged; only its certified norm budgets are sharpened. Repeated visits still refer to the same original sample. All unbounded multiplier identities below are polynomial quadratic-form identities; their finite compressions are self-adjoint finite matrices.

5.3.1 Activation and return with the improved variance

Lemma 5.6 (Fresh-sample contraction). *With $v_r^{(C)} = 4r + 4$, for $0 \leq k < \min(q, m)$,*

$$\|C : \mathcal{K}_k \rightarrow \mathcal{K}_{k+1}\| \leq \sqrt{(k+1)(m-k)v_r^{(C)}}. \quad (5.11)$$

The reverse map C^ has the same bound.*

Proof. 1. Compute the single-insertion Gram before the output cutoff.

For a sample currently in its local vacuum,

$$\begin{aligned} (Q_i \mathcal{X}_i P_i)^* (Q_i \mathcal{X}_i P_i) &= P_i \mathcal{X}_i Q_i \mathcal{X}_i P_i \\ &= P_i \mathcal{X}_i^2 P_i = (\mathbb{E} X_i^2) P_i \preceq v_r^{(C)} P_i. \end{aligned}$$

The middle equality uses $P_i \mathcal{X}_i P_i = 0$. It remains valid with arbitrary Hilbert-valued coefficients in the other sample registers. Output projection by Π_q can only decrease a norm.

2. Count the inputs that can meet at one output set.

Decompose $f = \sum_{|S|=k} f_S$ by exact active set; these inputs are orthogonal. For an output set T of size $k+1$, the contributing inputs are $f_{T \setminus \{i\}}$, one for each $i \in T$. Cauchy–Schwarz gives

$$\left\| \sum_{i \in T} \Pi_q Q_i \mathcal{X}_i P_i f_{T \setminus \{i\}} \right\|^2 \leq (k+1) v_r^{(C)} \sum_{i \in T} \|f_{T \setminus \{i\}}\|^2.$$

Sum over T . Each input set S appears once for each of the $m-k$ absent labels, yielding

$$\|Cf\|^2 \leq (k+1)(m-k) v_r^{(C)} \sum_{|S|=k} \|f_S\|^2.$$

Take square roots and then use adjointness. □

5.3.2 Wick input at a Hermite degree boundary

For one sample define the row multiplier

$$Tf = y^\top f, \quad T^*T = \mathcal{M}_Y$$

as a quadratic-form identity on polynomials. All following estimates allow a Hilbert-valued coefficient space for the other registers. Hölder is applied only in the current sample variables, with that coefficient space kept intact. Before the finite cutoff, identities involving multipliers and their adjoints are understood as polynomial quadratic-form identities; after compression the adjoints are ordinary finite-dimensional ones.

Lemma 5.7 (Local degree-dependent row contraction). *Let f_d have homogeneous local Hermite degree $d \geq 1$. For every integer $p \geq 2$,*

$$\|Tf_d\|_2^2 \leq M_p(r) \left(\frac{p+1}{p-1} \right)^d \|f_d\|_2^2. \quad (5.12)$$

For a finite sum $f = \sum_{d \geq 1} f_d$ with no local vacuum component, any specified choices of integers $p_d \geq 2$ give

$$\|Tf\|_2^2 \leq 3 \sum_{d \geq 1} M_{p_d}(r) \left(\frac{p_d+1}{p_d-1} \right)^d \|f_d\|_2^2. \quad (5.13)$$

Proof. **1. Apply Hölder on the original sample.**

The exponents are $2p$ and $2p/(p-1)$. Gaussian hypercontractivity (2.2) therefore gives

$$\begin{aligned} \|y^\top f_d\|_2 &\leq \| \|y\| \|f_d\| \|_{L_{2p}} \|f_d\|_{L_{2p/(p-1)}} \\ &\leq \| \|y\| \|_{L_{2p}} \left(\frac{p+1}{p-1} \right)^{d/2} \|f_d\|_2. \end{aligned}$$

Square and use (5.3) without relaxing $M_p(r)$. This proves (5.12). The multiplier and the polynomial use the same Gaussian sample; no independence between y and f_d is assumed.

2. Sum homogeneous inputs using their output-degree overlap.

Every coordinate of y is pure second chaos: it is a linear combination of $g_a h_b$ with coordinates in distinct Gaussian factors. Write $\mathcal{F}_e$ for exact local Hermite degree e , with the Hilbert coefficient space for the other registers left intact; set $\mathcal{F}_e = \{0\}$ when $e < 0$. Applying (4.2) twice shows

$$Tf_d \in \mathcal{F}_{d-2} \oplus \mathcal{F}_d \oplus \mathcal{F}_{d+2}.$$

At any output degree, at most three input degrees contribute. Cauchy–Schwarz there, followed by orthogonality across output degrees, yields $\|Tf\|_2^2 \leq 3 \sum_d \|Tf_d\|_2^2$. Apply (5.12) with the selected p_d separately. This proves (5.13). $\square$

A simple explicit exponent policy. Choosing $p_d = 2d + 1$ in (5.12) and using $M_{2d+1}(r) \leq (2d+1)(r+2d)$ gives

$$\|Tf_d\|_2^2 \leq e[(2d+1)r + 4d^2 + 2d] \|f_d\|_2^2, \quad (5.14)$$

since $(1 + 1/d)^d \leq e$. The factor three in (5.13) remains; the Wick improvement has not removed overlap or hypercontractivity. Define the simple repeated-visit allowance

$$\tilde{B}_q(r) = 27erq + 192eq^2 + 24eq. \quad (5.15)$$

No free λ is needed for this policy. Chain B retains its distinct $B_q(r; \lambda)$ in (4.11); it is not overwritten by $\tilde{B}_q$.

Lemma 5.8 (Sum of already-active sample interactions). *On the entire cutoff of (4.4),*

$$\|\mathcal{R}\| \leq \tilde{B}_q(r). \quad (5.16)$$

Its restriction to the vacuum sector is zero.

Proof. 1. Fix the exact active set and expose the centered form.

On an active-set sector S , with $k = |S| \leq q$,

$$\langle f, \mathcal{R}f \rangle = \sum_{i \in S} \|T_i f\|_2^2 - k \|f\|_2^2. \quad (5.17)$$

Indeed $Q_i f = f$ for $i \in S$ and $Q_i f = 0$ otherwise, and $\langle f, \mathcal{X}_i f \rangle = \|T_i f\|_2^2 - \|f\|_2^2$. In particular $\mathcal{R} \succeq -kI$ on this sector.

2. Sum each label's actual degree before relaxing any budget.

Let $f = \sum_{\mathbf{d}} f_{\mathbf{d}}$ be the joint local-degree decomposition. The number operators commute and their joint eigenspaces are orthogonal. Using (5.13) with the policy (5.14),

$$\begin{aligned} \sum_{i \in S} \|T_i f\|_2^2 &\leq 3e \sum_{\mathbf{d}} \left[r \left(2 \sum_{i \in S} d_i + k \right) + 4 \sum_{i \in S} d_i^2 + 2 \sum_{i \in S} d_i \right] \|f_{\mathbf{d}}\|_2^2 \\ &\leq 3e [9rq + 64q^2 + 8q] \|f\|_2^2 = \tilde{B}_q(r) \|f\|_2^2. \end{aligned}$$

The only budget relaxations are

$$\sum_i d_i \leq 4q, \quad k \leq q, \quad \sum_i d_i^2 \leq \left(\sum_i d_i \right)^2 \leq 16q^2.$$

Charging every active sample the total degree $4q$ separately would introduce an unnecessary extra factor of q ; summing the actual degrees avoids it.

3. Control both sides and preserve the cutoff.

Equation (5.17) gives $-kI \preceq \mathcal{R} \preceq \tilde{B}_q I$. Since $\tilde{B}_q \geq q \geq k$, its norm is at most $\tilde{B}_q$. These form inequalities survive the finite cutoff. Sum over orthogonal exact-support sectors; on $S = \emptyset$ the operator vanishes. $\square$

5.4 C4. Propagate the hybrid budgets and recover the endpoint

Put $K = \min(q, m)$ and retain the Wick activation coefficients

$$a_k^{(C)} = \frac{\sqrt{(k+1)(m-k)} v_r^{(C)}}{m}, \quad v_r^{(C)} = 4r + 4, \quad 0 \leq k < K. \quad (5.18)$$

Let $G_{C,q}$ be the nonnegative symmetric tridiagonal matrix with $(G_{C,q})_{00} = 0$, nonvacuum diagonal $\tilde{B}_q(r)/m$, and off-diagonal $a_k^{(C)}$. Activation, return, and repeated-visit estimates give $n(M_q f) \leq G_{C,q} n(f)$, so the vacuum identity yields

$$\mathbb{E} \operatorname{tr} Z^{2q} \leq r \|G_{C,q}^q e_0\|_2^2. \quad (5.19)$$

Here $n(f)$ is the vector of active-count sector norms from (4.16); it does not encode an independent replacement model.

Theorem 5.9 (Chain C closed-form moment and tail). *For every integer $q \geq 1$, with $c_q = ((2q-1)!!)^{1/(2q)}$ and $\eta_q = (r/\delta)^{1/(2q)}$,*

$$(\mathbb{E} \operatorname{tr} Z^{2q})^{1/(2q)} \leq r^{1/(2q)} \left[c_q \sqrt{\frac{4r+4}{m}} + \frac{\tilde{B}_q(r)}{m} \right], \quad (5.20)$$

$$\mathbb{P} \left\{ \|Z\| > \eta_q \left[c_q \sqrt{\frac{4r+4}{m}} + \frac{\tilde{B}_q(r)}{m} \right] \right\} \leq \delta. \quad (5.21)$$

Consequently Chain C also proves the common endpoint (1.2).

Proof. 1. Preserve the comparison vacuum.

As in the fully proved comparison in Theorem 4.5, $G_{C,q}$ is entrywise dominated by $bI + aJ_q$, with $b = \tilde{B}_q/m$ and $a = \sqrt{(4r+4)/m}$. The positive iteration preserves this domination. The scalar Hermite recurrence identifies $\|(bI + aJ_q)^q e_0\|^2$ with $\mathbb{E}(b + a\xi)^{2q}$, where ξ is standard Gaussian. This comparison random variable changes no original Gaussian sample.

2. Take the moment root and apply Markov.

Scalar Minkowski gives the first inequality. The spectral inequality $\|Z\|^{2q} \leq \text{tr } Z^{2q}$ and Markov give the second. With $q = \lceil \log(r/\delta) \rceil$, $\eta_q \leq \sqrt{e}$ and $c_q \leq \sqrt{2q}$, the allowance has order $\sqrt{rq/m} + rq/m + q^2/m + q/m$. Since $r, q \geq 1$ and $\varepsilon < 1$, requiring $m \geq C(\varepsilon^{-2}rt + \varepsilon^{-1}t^2)$, with $t = \log(4r/\delta)$, controls all terms. This is the same rate as A and B. $\square$

Stage-dependent hybrid propagation. For $1 \leq j \leq q$, let $G_C^{[j]}$ have the same off-diagonal $a_k^{(C)}$, zero vacuum diagonal, and nonvacuum diagonal $\tilde{B}_j(r)/m$. The exact prefix after $j-1$ steps belongs to the cutoff at total degree $4(j-1)$; its next multiplication needs only degree $4j$. Apply the C-contractions at that next cutoff and iterate sector norms in order:

$$\mathbb{E} \text{tr } Z^{2q} \leq r \|G_C^{[q]} \cdots G_C^{[1]} e_0\|_2^2 \leq r \|G_{C,q}^q e_0\|_2^2. \quad (5.22)$$

The proof is the same positive iteration as Theorem 4.7, with the *C-budgets* supplied by Lemmas 5.6 and 5.8; the operator identity is unchanged. The last inequality uses $\tilde{B}_j \leq \tilde{B}_q$. No stage matrices are commuted.

5.5 Refinement of C4: exact radial products and degree allocations

The simple C-stagewise bound (5.22) still uses AM–GM for $M_p(r)$, replaces each $(1 + 1/d)^d$ by e , and uses one diagonal allowance for every active count. We now retain those quantities. This is a finite deterministic refinement of the same operator proof, not a new random sketch or a simulation. Fix the terminal q and the finite exponent menu

$$\mathcal{P}_q = \{2, 3, \dots, 8q + 1\}.$$

For every even $d \in \{2, 4, \dots, 4q\}$ choose a fixed $p_d \in \mathcal{P}_q$ and put

$$h_d(r) = M_{p_d}(r) \left(\frac{p_d + 1}{p_d - 1} \right)^d, \quad b_d(r) = 3h_d(r) - 1. \quad (5.23)$$

Any fixed choices give valid local upper bounds. For the additional comparison with the simple certificate, we require

$$h_d(r) \leq M_{2d+1}(r) \left(\frac{d+1}{d} \right)^d. \quad (5.24)$$

For instance one may minimize over $\mathcal{P}_q$, since the menu contains $2d+1$. Alternatively a predetermined list satisfying (5.24) is sufficient; no proof that the list minimizes the menu is needed. The lists used in the numerical table are specified and checked in Appendix C.

For $1 \leq k \leq j \leq q$, define

$$\mathfrak{D}_{j,k}(r) = \max_{\substack{d_1, \dots, d_k \in 2\mathbb{N} \\ d_1 + \dots + d_k \leq 4j}} \sum_{i=1}^k b_{d_i}(r), \quad \mathfrak{D}_{j,0} = 0. \quad (5.25)$$

The symbol $\mathfrak{D}$ denotes this degree maximum, not the radial product $D_{r,p}$ in (5.1). All the costs use the same terminal choice of (p_d) throughout the q stages.

Lemma 5.10 (Exact-degree repeated-visit budget). *On the stage- j cutoff restricted to k active samples,*

$$\|\mathcal{R}|_{\mathcal{K}_k}\| \leq \mathfrak{D}_{j,k}(r).$$

For every admissible exponent choice $\mathfrak{D}_{j,k} \geq k$. If (5.24) also holds, then $\mathfrak{D}_{j,k} \leq \tilde{B}_j(r)$.

Proof. 1. Keep the centering term in each local cost.

On a fixed active set, apply (5.13) in (5.17) and then the joint degree decomposition. For every component the upper cost is $\sum_i (3h_{d_i} - 1) = \sum_i b_{d_i}$; the -1 terms sum to the exact $-k$ centering cost. The original symmetries require even $d_i \geq 2$ and the next cutoff requires total degree at most $4j$. This proves the stated upper form bound.

2. Bound the negative side and compare policies.

The negative form bound is $-kI$. Since $M_p(r) \geq r$ and the other factor in h_d is at least one, $h_d \geq r \geq 1$ and hence $b_d \geq 2$. Thus $\mathfrak{D}_{j,k} \geq k$, controlling both signs. Under (5.24), each h_d is at most the simple local allowance in (5.14). Sum as in Lemma 5.8 to get $\mathfrak{D}_{j,k} \leq \tilde{B}_j$. $\square$

Let $K = \min(q, m)$ and $a_k = \sqrt{(k+1)(m-k)v_r^{(C)}}/m$. Define $\hat{G}_C^{[j]}$ on $0, \dots, K$ by

$$\begin{aligned} (\hat{G}_C^{[j]})_{00} &= 0, \\ (\hat{G}_C^{[j]})_{kk} &= \begin{cases} \mathfrak{D}_{j,k}(r)/m, & 1 \leq k \leq \min(j, K), \\ 0, & j < k \leq K, \end{cases} \\ (\hat{G}_C^{[j]})_{k+1,k} &= (\hat{G}_C^{[j]})_{k,k+1} = a_k. \end{aligned} \tag{5.26}$$

The zero entries beyond the currently reachable active counts are padding.

Theorem 5.11 (Wick radial/degree vacuum certificate). *For the original independent Gaussian law and every fixed admissible exponent list,*

$$\mathbb{E} \operatorname{tr} Z^{2q} \leq r \|\hat{G}_C^{[q]} \hat{G}_C^{[q-1]} \cdots \hat{G}_C^{[1]} e_0\|_2^2. \tag{5.27}$$

In particular,

$$r \|\hat{G}_C^{[q]} \cdots \hat{G}_C^{[1]} e_0\|_2^2 \leq \delta \varepsilon^{2q} \tag{5.28}$$

is a sufficient original-law embedding certificate. Under (5.24) it is no worse than the simple stage-dependent certificate at the same q, r, m .

Proof. For the exact prefixes $f_j = \mathcal{M}_Z^j(e_a \otimes \Omega)$, the next cutoff identity is still $f_j = M_j f_{j-1}$ as in Theorem 4.7. Use Lemma 5.6 for activation and return, and Lemma 5.10 for the diagonal interaction. Hence $n(f_j) \leq \hat{G}_C^{[j]} n(f_{j-1})$ componentwise. Induct in stage order, square the terminal sector norm, and sum the r external vacuum seeds. This proves (5.27); Markov gives (5.28). Under (5.24), $\hat{G}_C^{[j]} \leq G_C^{[j]}$ entrywise, so positivity gives the claimed comparison. No commutation of the stage matrices is used. $\square$

5.6 Completion, scope, and finite-moment transfer

Either the simple or the refined C-certificate, followed by Markov, supplies the same arbitrary-fixed-subspace embedding guarantee. The common lower bound in Section 6 therefore matches C as well as A and B at fixed confidence and distortion. The constant $4r + 4$ is a uniform covariance envelope, not a sharp-covariance claim; the radial envelope is attained, but subsequent Hölder, degree-overlap, and scalar-majorant estimates can lose information.

The $2q$ -wise sample-block transfer of Corollary 4.9 applies to C: it transfers the *untruncated* polynomial moment $\mathbb{E} \operatorname{tr} Z^{2q}$. It neither constructs a finite-seed Gaussian generator nor transfers an inverse bound or the common lower-tail obstruction. As in B, no positive-coefficient cone, general descent theorem, or covariance-spectrum renewal theorem is needed here. The hybrid uses exact contractions at the local coefficient boundary, then the full-space Hermite framework.

6 A common matching lower bound

All three upper proofs concern the same distribution, so they share one lower-bound argument. A single large sample suffices; no spectral independence of matrix entries is required.

Theorem 6.1 (Common-factor obstruction). *Take $n_1 = r$, $n_2 = 2$, and $Ue_a = e_a \otimes e_1$ for $1 \leq a \leq r$. For all sufficiently large r , uniformly over integers $1 \leq m \leq \frac{1}{4}r \log r$,*

$$\mathbb{P}\{\|Z\| > 1/2\} \longrightarrow 1 \quad (r \longrightarrow \infty).$$

Proof. For this subspace,

$$y_i = \eta_i \xi_i, \quad \eta_i \sim N(0, 1), \quad \xi_i \sim N(0, I_r),$$

with all scalar and vector variables independent.

1. Below r samples, rank already forces failure.

If $m < r$, the sample covariance $m^{-1} \sum_i y_i y_i^\top$ has rank at most m , so it has a zero eigenvalue. Therefore $\|Z\| \geq 1$ deterministically.

2. Select the large scalar without biasing its vector factor.

Suppose $m \geq r$ and let J be the smallest index maximizing η_i^2 . The index depends only on the scalars. Conditional on all of them, ξ_J still has law $N(0, I_r)$; thus it is independent of those scalars. Positivity of the remaining covariance summands gives

$$\lambda_{\max} \left(\frac{1}{m} \sum_i y_i y_i^\top \right) \geq \frac{\eta_J^2 \|\xi_J\|^2}{m}. \quad (6.1)$$

3. Lower-bound the selected scalar and the vector length.

For $m \geq 3$, integrate the Gaussian density on $[a, a + 1/a]$ and its reflection, where $a = \sqrt{\log m} \geq 1$. Since $(a + 1/a)^2 \leq a^2 + 3$, this gives an absolute $c_1 > 0$ such that

$$p_m := \mathbb{P}\{\eta^2 \geq \log m\} \geq c_1 \frac{m^{-1/2}}{\sqrt{\log m}}.$$

Independence implies

$$\mathbb{P}\{\eta_J^2 < \log m\} = (1 - p_m)^m \leq \exp \left(-c_1 \frac{\sqrt{m}}{\sqrt{\log m}} \right).$$

Also $\mathbb{E} e^{-\|\xi\|^2/2} = 2^{-r/2}$, so exponential Markov gives

$$\mathbb{P}\{\|\xi_J\|^2 < r/2\} \leq e^{r/4} 2^{-r/2} \leq e^{-r/16}.$$

Consequently, with probability at least

$$1 - \exp \left(-c_1 \frac{\sqrt{m}}{\sqrt{\log m}} \right) - e^{-r/16}, \quad (6.2)$$

the right side of (6.1) is at least $r \log m / (2m)$.

4. Read off the necessary sample order.

If $r \leq m \leq \frac{1}{4}r \log r$, then $\log m \geq \log r$, and hence $r \log m / (2m) \geq 2$. On the event above, $\|Z\| \geq 1 > 1/2$. The probability in (6.2) tends to one uniformly throughout this range, since $m \geq r \rightarrow \infty$. Together with the rank argument this proves the claim. $\square$

At $\varepsilon = 1/2$, $\delta = 0.01$, one has $t = \log(400r) = O(\log(er))$ and $t^2 = O(r \log(er))$. Any of the three upper proofs and Theorem 6.1 therefore establish the $\Theta(r \log(er))$ assertion of Theorem 1.1. The obstruction is worst-case: a particular subspace may need fewer rows.

7 Comparing A, B, and C: endpoints and constants

7.1 Fix the labels before comparing a number

Chain A means the concentration proof of Section 3, including its original raw proxy $9r$ and moment scale $72(r+s)$. Chain B means the framework proof of Section 4, including $v_r = 9r - 1$ and $B_j(r; \lambda)$. Chain C means the Wick–Hermite proof of Section 5, with $v_r^{(C)} = 4r + 4$, $\tilde{B}_j$, and optionally $\mathfrak{D}_{j,k}$. *None of these labels changes meaning when a sharper estimate is discussed.*

For the principal comparison, define

$$U_A = \sqrt{\frac{18rt}{m}} + \frac{144(r+s)t}{m} + 3e^{-s}, \quad t = \log(4r/\delta), \quad s = \lceil \log(4m/\delta) \rceil; \quad (7.1)$$

$$U_B = \left[\frac{r}{\delta} \|G^{[q]} \cdots G^{[1]} e_0\|_2^2 \right]^{1/(2q)}; \quad (7.2)$$

$$U_C = \left[\frac{r}{\delta} \|\hat{G}_C^{[q]} \cdots \hat{G}_C^{[1]} e_0\|_2^2 \right]^{1/(2q)}. \quad (7.3)$$

In the cover diagram $G_B^{[j]}$ is simply an explicit alias for B’s $G^{[j]}$ in (4.22). The C matrices in (5.26) are different majorants of the same underlying multiplication. Each displayed allowance being at most ε certifies $\mathbb{P}\{\|Z\| > \varepsilon\} \leq \delta$.

7.2 The local constants, side by side

Quantity	A: concentration	B: framework	C: Wick–Hermite
Row input at L_{2p} , squared	$r(4p-3) + (2p-2)^2$	Same baseline	$M_p(r) = (\kappa_p D_{r,p})^{1/p} \leq p(r+p-1)$
Raw second moment	$9r$	$9r$	$4r+5$
Centered activation variance	Not the assembly parameter	$9r-1$	$4r+4$
Exponential-moment scale	$72(r+s)$	Not used	Not used
Simple repeat allowance	Not used	$99erq + 768eq^2$ at $\lambda = 2$	$27erq + 192eq^2 + 24eq$
Stronger propagation	Matrix Laplace	Ordered $G^{[q]} \cdots G^{[1]}$	Ordered $\hat{G}_C^{[q]} \cdots \hat{G}_C^{[1]}$; active-degree maximum
Untruncated trace moments	Not supplied by its displayed tail certificate	Yes	Yes

Table 2: The fixed A/B baselines and the separate C refinement. The numerical inequalities are proved in their own sections; shared symbols do not overwrite old constants.

For B also retain the closed and stationary allowances

$$U_{B,\text{cl}} = \eta_q \left[c_q \sqrt{\frac{9r-1}{m}} + \frac{B_q(r; \lambda)}{m} \right], \quad (7.4)$$

$$U_{B,\text{st}} = \left[\frac{r}{\delta} \|G_q^q e_0\|_2^2 \right]^{1/(2q)}. \quad (7.5)$$

At identical m, r, q, λ, δ , $U_B \leq U_{B,\text{st}} \leq U_{B,\text{cl}}$. For C denote the allowance from (5.22) by $U_{C,\text{simple}}$. Under (5.24), $U_C \leq U_{C,\text{simple}}$ at identical parameters. For the comparison $\lambda = 2$, $\tilde{B}_j \leq B_j(r; 2)$ because their difference is $72erj + 576ej^2 - 24ej > 0$. Also $4r + 4 \leq 9r - 1$ for $r \geq 1$. Thus $U_C \leq U_{C,\text{simple}} \leq U_B$ at the same parameters and that baseline policy. The $r = 10$ table uses $\lambda = 1$; there the analogous budget inequality is checked for the displayed stages. No universal comparison with every optimized concentration proof is inferred.

7.3 The main numerical comparison, with proof parameters

Table 1 is the principal A/B/C comparison. The model and $\varepsilon = 1/2, \delta = 0.01$ are identical. B and C use the same terminal q , so the reduction from B to C is not obtained by changing the moment order. B uses $\lambda = 1$ at $r = 10$ and $\lambda = 2$ otherwise. C’s exact exponent lists appear in Table 5.

r	q	λ_B	A	B	C
10	3	1	91 000	60 000	12 800
100	5	2	580 000	472 000	111 000
1 000	7	2	6 200 000	5 150 000	1 250 000
10 000	8	2	72 000 000	61 000 000	15 000 000

Table 3: All parameters of the primary A/B/C certificates; C uses the specified radial/degree policy, not an unspecified optimization.

The comparison reduces the displayed B row counts by factors approximately 4.69, 4.25, 4.12, and 4.07, respectively. These ratios concern the selected certificates, not the true minimum sample complexity. In particular, A and B are intentionally the original baselines: their separate identity is what makes the contribution of C visible.

7.4 The gains and losses behind the constants

Let $h = \log(r/\delta)$ and $q = \lceil h \rceil$, with $h \rightarrow \infty$. Stirling gives $c_q \sim \sqrt{2q/e}$ and $\eta_q \sim \sqrt{e}$. Consequently the fresh-sample term has asymptotic coefficient

$$\text{A and B: } \sqrt{18rh/m}, \quad \text{C: } \sqrt{8rh/m},$$

when also $r \rightarrow \infty$. The improved covariance envelope alone reduces that coefficient by a factor $3/2$. At $\lambda = 2$ and $r/h \rightarrow \infty$, the simple repeated-visit coefficient in the closed form changes from $99e^{3/2} \approx 443.69$ for B to $27e^{3/2} \approx 121.01$ for C; the exact-degree certificate can improve further. Neither coefficient is a globally optimal constant.

Retaining the vacuum rather than replacing it by Schur’s full-norm bound still avoids the asymptotic fresh-term factor $\sqrt{2e}$ in either framework route. C has *not* removed the three-degree-overlap factor, hypercontractivity on arbitrary polynomial coefficients, or the block-norm majorization itself. Its sharp radial bound does not imply an optimal final embedding constant.

Same endpoint, a distinguishable hybrid

A, B, and C all yield $O(\varepsilon^{-2}r \log(4r/\delta) + \varepsilon^{-1} \log^2(4r/\delta))$, and the common fixed-confidence lower bound gives $\Theta(r \log(er))$. The difference is the proved coefficient budget and the resulting explicit certificate. C has a separate derivation and improves the stated A/B baselines. Appendix D records a fair additional comparison after applying Wick inputs to A as well; it prevents interpreting this table as intrinsic superiority of one general method.

8 Scope, provenance, and verification status

The factors throughout are independent real standard Gaussian vectors, and the subspace is deterministic. No step establishes the same improvement for arbitrary isotropic sub-Gaussian factor vectors, higher tensor orders, or subspaces chosen after observing the sketch. The lower bound is worst-case, not a classification of every subspace by Schmidt geometry. Optimal joint ε, δ dependence is not claimed.

Chain A reflects the coefficient-retention viewpoint but is a standalone concentration argument. Chains B and C use the framework’s exact multiplication, graded-domain reduction, coefficient-sensitive block estimates, and vacuum/energy propagation. Neither needs the positive-coefficient cone, the graph connected-creation theorem, general encoding/descent tests, signed conditional cores, or covariance-spectrum renewal. In particular, their Gaussian hypercontractivity bounds apply to arbitrary polynomial coefficients.

The present edition corrects the preceding integration’s organization: it restores A and B as the original baselines and gives the Wick–Hermite refinement its own Chain C. The supplied three-chain rMPS manuscript is a layout reference only. All theorem statements and numerical quantities here concern the same Khatri–Rao embedding model. The prior operator-domain correction and expanded proof exposition are retained.

The starting material was the two proof notes developed in this conversation: *Sharp fixed-confidence order-two Gaussian Khatri–Rao embeddings* and *Gaussian Khatri–Rao embeddings through the graded multiplication pipeline*, both dated 16 September 2026. This document preserves their baseline application arguments and integrates the Wick supplement as the separate Chain C [23, 22, 21]. The organizing framework is the supplied v11 manuscript [1]. The Gaussian inequalities, moment-estimation methods, and matrix-analysis inputs are classical and are credited at their points of use. Local proof provenance in a conversation is not evidence of publication priority. In particular, the row-moment order has the prior implication explained after Lemma 2.2, and Chain A’s general proof strategy has the predecessors recorded in Section 1.3.

8.1 Attribution by ingredient

Ingredient	Prior source or standard step	Role of the present note
Sketch construction and earlier guarantees	Tensor random projection [2]; fixed-vector sketching [3]; rank-one probing [4]; earlier OSE analysis [5, 8].	Same sketch, with the precise real-Gaussian, order-two theorem proved here. No claim to introduce the sketch.
Gaussian functional inequalities	Nelson hypercontractivity; Gross logarithmic Sobolev theory; Aida–Stroock moment methods [9, 10, 11, 12].	Re-derivations fix constants and the Hilbert-valued normalization.
Projected-row moments	The order follows from the Hilbert case of [13, Corollary 12]. Fourth moments are Isserlis calculations [14].	Explicit baseline row bound and its use in A/B; not a new general chaos-moment order.
Concentration chain	Khatri–Rao truncation strategy [8], with the version-sensitive predecessor [6, 7]; Lieb–Tropp matrix Laplace/Bernstein [15, 16].	Rank-one matrix-moment bookkeeping, all-order cutoff calculation, bias control, and one explicit certificate.
Graded representation	Classical chaos, probability-law representations, and Jacobi moments [17, 18, 19]; organization in [1].	Exact support/degree restriction for this model. Representation alone is not claimed as novel.
Coefficient and energy estimates	Coefficient-sensitive chaos methods [13, 20]; block norms and positive majorization; energy interface in [1].	Specific activation, local-degree repeated-visit, and ordered vacuum calculations with displayed constants.
Wick–Hermite hybrid	Classical Gaussian pairings and chaos [14, 21]; radialization recorded in [22]; supplied calculations in [23].	Sharp radial product, orthogonal mixed-fourth estimate, and refined C-stage budgets proved here. Internal provenance is not external priority.
Lower bound and moment transfer	Elementary positivity, Gaussian tails, and low-degree moment matching.	Explicit common-factor witness and sample-block transfer proof; no claim to invent these general proof techniques or a finite-seed construction.

The numerical comparison is internal: all constants come from the implementations proved here, not from an optimization over the cited authors’ theorems. Showing that one displayed certificate is smaller than another establishes no universal ranking of the methods. The source checks identify relevant antecedents but are not an exhaustive literature search establishing originality of the final embedding theorem.

8.2 Verification status

The accompanying checks verify exact finite scalar inequalities, selected high-precision numerical certificates, and normalization benchmarks. They do not constitute an independent proof audit, a Lean certificate, a test of every possible parameter value, or a claim of publication priority. The general statements are justified by the analytic proofs, not by the numerical tests.

A Gaussian inputs and Hermite normalization

This appendix records short standard semigroup arguments for the Gaussian inequalities used in Section 2. They are included to fix constants, not asserted as new results; see Nelson [9] and Gross [10, 11].

A.1 Ornstein–Uhlenbeck semigroup and logarithmic Sobolev inequality

For a smooth function F on $\mathbb{R}^N$, let

$$(\mathbf{P}_t F)(x) = \mathbb{E} F(e^{-t}x + \sqrt{1 - e^{-2t}}G), \quad G \sim N(0, I_N).$$

Standard Gaussian measure is invariant under $\mathbf{P}_t$; this follows directly because a sum of independent centered Gaussian vectors with squared coefficients summing to one is standard Gaussian. Differentiation of the kernel gives the generator $\mathcal{L} = \Delta - x \cdot \nabla$, and Gaussian integration by parts gives

$$\mathbb{E} f \mathcal{L} g = -\mathbb{E} \langle \nabla f, \nabla g \rangle, \quad \nabla \mathbf{P}_t F = e^{-t} \mathbf{P}_t \nabla F.$$

1. Differentiate the entropy along the semigroup.

Assume first that F is smooth, bounded, and bounded below by a positive constant. Set $u_t = \mathbf{P}_t F$. Since $\mathbb{E} u_t = \mathbb{E} F$,

$$\frac{d}{dt} \text{Ent}(u_t) = \mathbb{E}(\mathcal{L} u_t) \log u_t = -\mathbb{E} \frac{\|\nabla u_t\|^2}{u_t}.$$

As $t \rightarrow \infty$, $u_t \rightarrow \mathbb{E} F$ and its entropy tends to zero, so integrating this derivative recovers $\text{Ent}(F)$.

2. Bound the dissipation pointwise.

The gradient commutation identity and weighted Cauchy–Schwarz give pointwise

$$\frac{\|\nabla \mathbf{P}_t F\|^2}{\mathbf{P}_t F} \leq e^{-2t} \mathbf{P}_t \left(\frac{\|\nabla F\|^2}{F} \right).$$

3. Integrate in space, then in time.

Integrate in Gaussian measure, use invariance, and then integrate over t :

$$\text{Ent}(F) \leq \int_0^\infty e^{-2t} dt \mathbb{E} \frac{\|\nabla F\|^2}{F} = \frac{1}{2} \mathbb{E} \frac{\|\nabla F\|^2}{F}.$$

Taking $F = f^2$ gives

$$\text{Ent}(f^2) \leq 2 \mathbb{E} \|\nabla f\|^2,$$

which is (2.1). Positive regularization, truncation, and smooth approximation extend the inequality to the functions with finite energy used in the main proof.

A.2 Hypercontractivity and homogeneous chaos

1. A monotone norm along the semigroup.

Let $f > 0$ be smooth and bounded, $u_t = \mathbf{P}_t f$, and $p(t) = 1 + (p_0 - 1)e^{2t}$ with $p_0 > 1$. Differentiation yields

$$\frac{d}{dt} \log \|u_t\|_{L_{p(t)}} = \frac{p'(t)}{p(t)^2} \frac{\text{Ent}(u_t^{p(t)})}{\mathbb{E} u_t^{p(t)}} - (p(t) - 1) \frac{\mathbb{E} u_t^{p(t)-2} \|\nabla u_t\|^2}{\mathbb{E} u_t^{p(t)}}.$$

Apply (2.1) to $u_t^{p(t)/2}$:

$$\text{Ent}(u_t^{p(t)}) \leq \frac{p(t)^2}{2} \mathbb{E} u_t^{p(t)-2} \|\nabla u_t\|^2.$$

Since $p'(t) = 2(p(t) - 1)$, the derivative of the logarithmic norm is nonpositive. Thus $\|P_t f\|_{L_{p(t)}} \leq \|f\|_{L_{p_0}}$. Positivity of the semigroup and $|P_t f| \leq P_t |f|$ extend this to signed functions, then approximation extends it to the relevant L_p spaces.

2. Specialize to homogeneous chaos.

The Hermite generating function

$$e^{zx - z^2/2} = \sum_{\ell=0}^{\infty} \text{He}_{\ell}(x) \frac{z^{\ell}}{\ell!}$$

shows that P_t acts on homogeneous total degree d by e^{-dt} . Take $p_0 = 2$ and $u = 1 + e^{2t}$. For a scalar homogeneous Hermite polynomial F_d ,

$$e^{-dt} \|F_d\|_{L_u} \leq \|F_d\|_{L_2}, \quad \text{hence} \quad \|F_d\|_{L_u} \leq (u - 1)^{d/2} \|F_d\|_{L_2}.$$

3. Allow Hilbert-space coefficients.

For a Hilbert-valued polynomial with orthonormal coordinates $F_{d,a}$, Minkowski in $L_{u/2}$ gives

$$\|F_d\|_{L_u}^2 = \left\| \sum_a |F_{d,a}|^2 \right\|_{L_{u/2}} \leq \sum_a \|F_{d,a}\|_{L_u}^2 \leq (u - 1)^d \sum_a \|F_{d,a}\|_{L_2}^2.$$

Finite-dimensional projections and monotone convergence justify arbitrary Hilbert coefficient spaces. This proves (2.2) with the stated constant.

A.3 The scalar Jacobi comparison is exact through the needed moment

For $h_{\ell} = \text{He}_{\ell} / \sqrt{\ell!}$, the generating function also yields (4.2). The first $q + 1$ Hermite functions therefore represent compressed scalar multiplication by $\xi \sim N(0, 1)$ with Jacobi matrix J_q . An induction on the prefix length gives

$$(bI + aJ_q)^j e_0 \longleftrightarrow (b + a\xi)^j, \quad 0 \leq j \leq q.$$

The Gaussian moment recurrence, obtained by integration by parts, is $\mathbb{E} \xi^{2q} = (2q - 1) \mathbb{E} \xi^{2q-2}$. It follows that

$$\|J_q^q e_0\|^2 = (2q - 1)!!, \quad \|(bI + aJ_q)^q e_0\|^2 = \mathbb{E}(b + a\xi)^{2q}.$$

These are vacuum identities, not traces of the entire Jacobi matrix. They are the Hermite case of the classical Jacobi/Gaussian-quadrature correspondence [19]; the Gaussian moment recurrence is also the one-variable case of the product-moment formula [14].

B An all-parameter verification of the constant 72

We prove (3.3) without optimizing the constant. Set

$$a_p = 8p - 11, \quad b_p = (4p - 6)^2, \quad F_p(r) = \frac{(a_p r + b_p)^{p-1}}{r(r + p)^{p-2}}.$$

For integer $p \geq 3$, the sign of the logarithmic derivative in r is the sign of

$$[-8p^3 + 37p^2 - 62p + 36]r - b_p p.$$

Indeed, multiply $(p-1)a_p/(a_p r + b_p) - 1/r - (p-2)/(r+p)$ by its positive common denominator and collect terms. If $z = p-3 \geq 0$, the bracket becomes

$$-8z^3 - 35z^2 - 56z - 33 < 0.$$

Thus $F_p(r)$ decreases for $r \geq 1$, so it is enough to prove the inequality at $r = 1$. Since $a_p + b_p = (4p-5)^2$, the ratio of its left to right sides is

$$\rho_p = \frac{2(4p-5)^{2p-2}}{3p! 72^{p-2}(p+1)^{p-2}}. \quad (\text{B.1})$$

Finite orders $3 \leq p \leq 10$.

The following rational upper bounds are verified by integer cross-multiplication in (B.1); each is less than one.

p	3	4	5	6	7	8	9	10
Upper bound on ρ_p	927/1000	380/1000	177/1000	88/1000	46/1000	25/1000	14/1000	8/1000

For example $\rho_3 = 2401/2592 < 927/1000$. All eight exact fractions are also recorded by the companion verification script.

All remaining orders $p \geq 11$.

The elementary inequality $p! \geq (p/e)^p$ follows from $\sum_{k=1}^p \log k \geq \int_1^p \log x \, dx$. Using $4p-5 \leq 4p$ and $p+1 \geq p$ in (B.1) gives

$$\rho_p \leq \frac{32e^2}{3} \left(\frac{2e}{9} \right)^{p-2}.$$

Since $e < 11/4$ and $11/18 < 1$, for every $p \geq 11$,

$$\rho_p < \frac{242}{3} \left(\frac{11}{18} \right)^9 = \frac{285311670611}{297538935552} < 1.$$

This proves (3.3) for every real $r \geq 1$ and integer $p \geq 3$. It is an all-parameter analytic certificate with only eight finite integer checks, not a numerical sweep over dimensions or high moment orders.

C Numerical certificates and reproducibility

C.1 One recurrence, with explicit chain-dependent coefficients

For $K = \min(q, m)$ use

$$a_k^{(B)} = \frac{\sqrt{(k+1)(m-k)(9r-1)}}{m}, \quad a_k^{(C)} = \frac{\sqrt{(k+1)(m-k)(4r+4)}}{m}.$$

For B take $b_{j,k} = B_j(r; \lambda)$ for $k \geq 1$. For simple C take $b_{j,k} = \tilde{B}_j(r)$; for refined C take $b_{j,k} = \mathfrak{D}_{j,k}(r)$ when $1 \leq k \leq j$ and zero for padding. For stationary propagation replace j by q in its simple budget. In every case $b_{j,0} = 0$. Start from $w^{(0)} = e_0$ and update

$$w_k^{(j)} = a_{k-1} w_{k-1}^{(j-1)} + \frac{b_{j,k}}{m} w_k^{(j-1)} + a_k w_{k+1}^{(j-1)}. \quad (\text{C.1})$$

Out-of-range terms are omitted. Select a_k from the same chain as the diagonal budget; do not mix B and C inadvertently. The failure certificate is

$$F = \frac{r}{\varepsilon^{2q}} \sum_{k=0}^K (w_k^{(q)})^2. \quad (\text{C.2})$$

After diagonal costs are precomputed, the recurrence takes $O(qK)$ arithmetic operations and $O(K)$ storage. Positive interval arithmetic encloses it without randomized simulation.

C.2 Computing the exact degree maximum

For costs $b_d(r)$ from (5.23), let $F_{\text{deg}}(k, n)$ be the maximum sum for k active labels of exact half-degree n . Set $F_{\text{deg}}(0, 0) = 0$ and omit infeasible states. Then

$$\begin{aligned} F_{\text{deg}}(k, n) &= \max_{1 \leq \ell \leq n} \{F_{\text{deg}}(k-1, n-\ell) + b_{2\ell}(r)\}, \\ \mathfrak{D}_{j,k}(r) &= \max_{k \leq n \leq 2j} F_{\text{deg}}(k, n). \end{aligned} \tag{C.3}$$

For a fixed terminal q only degrees up to $4q$ occur. A direct implementation costs $O(q^3)$ operations to precompute this finite maximum and $O(q^2)$ storage. This maximum is over deterministic degree budgets, not over simulated samples.

C.3 Specified exponent choices

The following table lists p_d in order $d = 2, 4, \dots, 4q$. Each choice is fixed before interval evaluation. The supplementary checks verify (5.24) for each listed degree; claiming a global optimal exponent is neither needed nor intended.

r	q	$p_2, p_4, \dots, p_{4q}$
10	3	4, 6, 9, 11, 13, 16
100	5	4, 8, 11, 15, 18, 21, 24, 27, 30, 33
1 000	7	5, 8, 12, 16, 20, 24, 28, 31, 35, 39, 43, 46, 50, 54
10 000	8	5, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64

Table 5: Admissible exponent lists used in the refined C-column.

The finite menu $\mathcal{P}_q$ is only a convenient policy. The theorem does not require minimization over this menu; a chosen list always gives a valid certificate, while (5.24) supplies the additional comparison with the simple policy.

C.4 Verified allowances in the main A/B/C table

Each entry below is rounded upward to six decimal places at its row count in Table 1; all are strictly below $1/2$.

r	A	B	C
10	0.495576	0.494373	0.499783
100	0.497052	0.498858	0.499382
1 000	0.499705	0.499417	0.499353
10 000	0.499744	0.499431	0.498662

Table 6: Twelve main-table distortion certificates. Computed from proved inequalities, not estimated from random samples.

The companion `khatri_rao_three_chain_checks.py` uses 80-digit evaluation and 70-digit outward-rounded intervals for the twelve principal certificates, eight supplementary certificates in Appendix D, the fixed exponent policies, and their degree budgets. It records the chain identity explicitly in the output. All finite exponent choices are fixed before interval validation; no claim that a selected exponent is globally optimal is needed. The original verification scripts and outputs are included as archived checks with their original A/B labels; the new manifest maps them to the present chains. The analytic arguments, not the finite checks, justify the theorem for arbitrary subspaces and all admissible parameters.

C.5 Exact scalar and normalization checks

For the scalar common-factor model, let $X = g^2 h^2 - 1$. With $(-1)!! = 1$,

$$\mu_k := \mathbb{E}X^k = \sum_{\ell=0}^k \binom{k}{\ell} (-1)^{k-\ell} [(2\ell-1)!!]^2. \quad (\text{C.4})$$

If $M_k^{(j)} = \mathbb{E}(X_1 + \cdots + X_j)^k$, independence yields the exact recurrence

$$M_k^{(j+1)} = \sum_{\ell=0}^k \binom{k}{\ell} M_{k-\ell}^{(j)} \mu_\ell, \quad M_0^{(0)} = 1, \quad M_k^{(0)} = 0 \quad (k > 0).$$

Dividing $M_{2q}^{(m)}$ by m^{2q} gives an exact rational scalar trace moment. The script checks it against the stage-dependent majorant for $m \in \{1, 2, 5\}$ and $1 \leq q \leq 6$, giving 18 checks. At $q = 1$, both sides equal $8/m$ exactly because $\mathbb{E}X^2 = 9 - 1 = 8$. It separately checks $\|J_q^q e_0\|^2 = (2q-1)!!$ for $1 \leq q \leq 8$.

The archived baseline run verifies 16 interval sample certificates, 18 exact scalar-moment comparisons, eight Jacobi normalizations, eight finite integer inequalities for the moment scale, and one all-orders rational envelope. The scalar tests are useful indexing and normalization checks; they are not proofs for arbitrary subspaces. That generality comes from the analytic coefficient and operator estimates in the main text.

C.6 Non-scalar Wick checks and their limits

The Wick script computes rational polynomial expectations on four coefficient families: a two-dimensional common-factor subspace, the first two or three columns of the rational Householder matrix

$$H = I_4 - \frac{2uu^\top}{30}, \quad u = (1, 2, 3, 4)^\top,$$

reshaped as 2×2 matrices, and the full 2×2 tensor space. It checks 24 radial inequalities (orders 2 through 12), four exact covariance Loewner inequalities, and eleven orthogonal mixed-fourth inequalities. The common-factor radial moments attain (5.2) exactly.

For twelve centered fourth-moment comparisons it uses the exact ordered independent-label expansion

$$\mathbb{E} \operatorname{tr} Z^4 = \frac{\mathbb{E} \operatorname{tr} X^4 + (m-1)[2 \operatorname{tr} V^2 + \mathbb{E} \operatorname{tr}(X_1 X_2 X_1 X_2)]}{m^3}, \quad V = \mathbb{E}X^2.$$

Only the all-equal label and the three pair-pair patterns survive centering; the displayed alternating word is retained in its original order. This is an elementary independent-label identity, not a new cumulant theorem. The exact rational moments are compared numerically with the analytic C-majorant. These examples check indexing, normalization, and some non-common-factor geometry; they do not prove a statement for every subspace. That generality comes from the analytic operator and coefficient arguments.

D A supplementary Wick-assisted concentration variant

Not a fourth main chain and not the meaning of Chain C

The three principal chains are A, B, and C. This appendix separately records A^W : the same concentration architecture as A supplied with C’s better Gaussian inputs. It preserves the earlier scale-eight result and makes the constants comparison fair, without silently redefining Chain A.

This variant uses the radial and covariance estimates proved in Section 5 but *not* its Hermite propagation. The concentration step remains Lieb–Tropp; no new matrix concentration theorem is asserted. Write $\nu_r^{(C)} = 4r + 5$ for the raw matrix-moment proxy.

D.1 The Wick-assisted one-row moment estimate

Lemma D.1 (Wick-refined one-row matrix moments). *For every integer $p \geq 2$,*

$$\mathbb{E}Y^p \preceq \frac{p!}{2} \nu_r^{(C)} [8(r+p)]^{p-2} I_r. \quad (\text{D.1})$$

For $p \geq 3$ the sharper intermediate bound is

$$\mathbb{E}Y^p \preceq 3\sqrt{\kappa_{2p-2} D_{r,2p-2}} I_r. \quad (\text{D.2})$$

Proof. 1. Retain the rank-one matrix direction.

For unit x , $Y^p = \|y\|^{2p-2} yy^\top$ and Cauchy–Schwarz give

$$x^\top (\mathbb{E}Y^p) x \leq (\mathbb{E}\|y\|^{4p-4})^{1/2} (\mathbb{E}\langle x, y \rangle^4)^{1/2} \leq 3\sqrt{\kappa_{2p-2} D_{r,2p-2}}.$$

This is (D.2). The directional factor is bounded by Lemma 2.3. The case $p = 2$ of (D.1) is Corollary 5.4.

2. Bound the scalar pairing factor by a factorial sequence.

Write $H_p = \sqrt{\kappa_{2p-2}} = \sqrt{(4p-5)!!}$. For $p \geq 3$,

$$H_3 = \sqrt{105}, \quad \frac{H_{p+1}}{H_p} = \sqrt{(4p-3)(4p-1)} < 4p < 4(p+1).$$

Induction yields

$$H_p \leq \frac{\sqrt{105}}{24} p! 4^{p-2}. \quad (\text{D.3})$$

No finite parameter sweep is being substituted for an all-order argument.

3. Keep the first two radial factors, then use AM–GM.

There are $2p - 4$ remaining factors after $r(r+2)$, and their arithmetic mean is $r+2p-1$. Thus

$$\sqrt{D_{r,2p-2}} \leq \sqrt{r(r+2)}(r+2p-1)^{p-2} \leq \sqrt{r(r+2)}[2(r+p)]^{p-2}.$$

Together with (D.3), this bounds (D.2) by

$$\frac{\sqrt{105}}{8} p! \sqrt{r(r+2)} [8(r+p)]^{p-2} I_r.$$

Finally $\sqrt{105}/4 < 3$, $\sqrt{r(r+2)} \leq r+1$, and $3(r+1) \leq 4r+5$. The coefficient is at most $(p!/2)\nu_r^{(C)}$, proving (D.1). $\square$

The matrix direction is retained until expectation; the estimate is on $\|\mathbb{E}Y^p\|$, not merely $\mathbb{E}\|Y\|^p$. The constant eight is a uniform factorial envelope, not an asserted optimal scale. Appendix B proves the constant-72 calculation used by the unchanged Chain A.

D.2 Auxiliary truncation and all-order moments

The untruncated Y need not have a finite positive exponential moment. For example, when $r = 1$ and $y = gh$, $Y = g^2 h^2$ has infinite moment-generating function at every positive argument. Conditional on g , the Gaussian integral in h already diverges whenever $2\theta g^2 \geq 1$, an event of positive probability. Thus a positive-matrix exponential theorem cannot be applied to Y directly.

Lemma D.2 (All-order moments of a bounded auxiliary matrix). *Fix an integer $s \geq 2$ and define*

$$R_s = 8(r + s), \quad L_s = sR_s, \quad W = Y\mathbf{1}_{\{\|y\|^2 \leq L_s\}}, \quad \Sigma = \mathbb{E}W. \quad (\text{D.4})$$

Then

$$\mathbb{P}\{W \neq Y\} \leq e^{-2s}, \quad (\text{D.5})$$

$$0 \preceq I_r - \Sigma \preceq 3e^{-s}I_r, \quad (\text{D.6})$$

$$\mathbb{E}W^p \preceq \frac{p!}{2} \nu_r^{(C)} R_s^{p-2} I_r \quad \text{for every integer } p \geq 2. \quad (\text{D.7})$$

Proof. **1. Bound the event on which the sample is changed.**

At exponent $2s$, Theorem 5.1 gives

$$\|\|y\|^2\|_{L_s} \leq M_s(r) \leq s(r + s - 1).$$

Markov's inequality therefore gives

$$\mathbb{P}\{\|y\|^2 > L_s\} \leq \left(\frac{s(r + s - 1)}{8s(r + s)} \right)^s \leq 8^{-s} \leq e^{-2s}.$$

2. Charge the covariance bias in each direction.

For a unit vector x , Lemma 2.3 and Cauchy–Schwarz imply

$$x^\top (I_r - \Sigma)x = \mathbb{E}[\langle x, y \rangle^2 \mathbf{1}_{\{\|y\|^2 > L_s\}}] \leq 3\mathbb{P}\{\|y\|^2 > L_s\}^{1/2} \leq 3e^{-s}.$$

The integrand is nonnegative, proving both sides of (D.6).

3. Prove the matrix moment bound through order s .

For $p = 2$, $\mathbb{E}W^2 \preceq \mathbb{E}Y^2 \preceq \nu_r^{(C)} I_r$. For $3 \leq p \leq s$, the scalar event in (D.4) gives $W^p \preceq Y^p$ pointwise. Lemma D.1 and $p \leq s$ yield

$$\mathbb{E}W^p \preceq \frac{p!}{2} \nu_r^{(C)} [8(r + p)]^{p-2} I_r \preceq \frac{p!}{2} \nu_r^{(C)} R_s^{p-2} I_r.$$

4. Extend to every order required by the exponential series.

For an integer $p > s$, spectral calculus on $0 \preceq W \preceq L_s I_r$ gives

$$W^p \preceq L_s^{p-s} W^s.$$

Consequently,

$$\mathbb{E}W^p \preceq \frac{s!}{2} \nu_r^{(C)} R_s^{s-2} L_s^{p-s} I_r = \frac{s! s^{p-s}}{2} \nu_r^{(C)} R_s^{p-2} I_r \preceq \frac{p!}{2} \nu_r^{(C)} R_s^{p-2} I_r.$$

The last inequality follows by comparing each factor of $(s+1)(s+2)\cdots p$ with s . This proves all of (D.7), not just the first s moments. $\square$

The almost-sure cutoff height is $L_s = sR_s$, but the matrix moment-growth scale is only R_s . Charging L_s as the bounded-summand scale would lose the factor s .

D.3 Restore the original samples and compare with C

Apply the concentration lemma already proved in Chain A, Lemma 3.4, with $v = 4r + 5$ and $R = 8(r + s)$. Choose $t = \log(4r/\delta)$ and $s = \lceil \log(4m/\delta) \rceil$. The all-order estimate above supplies its hypotheses. A union bound for clipping gives at most $me^{-2s} \leq \delta/16$; the concentration event costs $2re^{-t} = \delta/2$. The covariance bias is at most $3e^{-s}$. On the good event the original samples equal the auxiliary ones, so

$$\mathbb{P} \left\{ \|Z\| > U_{A^W} := \sqrt{\frac{2(4r+5)t}{m}} + \frac{16(r+s)t}{m} + 3e^{-s} \right\} \leq \delta. \quad (\text{D.8})$$

The sample-order absorption proved in Proposition 3.6 applies with these smaller constants. This is still the original Gaussian sketch, not a clipped estimator.

r	Supplementary A^W	C simple	C refined
10	13 200	19 700	12 800
100	98 100	148 000	111 000
1 000	1 100 000	1 620 000	1 250 000
10 000	12 800 000	19 300 000	15 000 000

Table 7: A fair supplementary comparison after giving the concentration route the Wick input as well. The refined C-column repeats the primary C certificate; A in the main table continues to mean the original concentration baseline.

The corresponding upward-rounded allowances for A^W are 0.499192, 0.499736, 0.497858, 0.498718; those for simple C are 0.499266, 0.498479, 0.499814, 0.499373. The same q values as in Table 1 are used in both C-columns. All entries are sufficient certificates, not global minima. The concentration variant is smaller than refined C in three rows, while refined C is smaller at $r = 10$. The large improvement of C over the original A/B baselines is therefore not evidence of universal superiority over concentration.

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